LABORATORY MANUAL

Internet of Things

B. Tech - V Semester

(Skill Oriented Lab)
(For R-2020 Regulations)

COLLEGE OF ENGINEERING
(AUTONOMOUS)

DEPARTMENT OF MECHANICAL ENGINEERING

GAYATRI VIDYA PARISHAD COLLEGE OF ENGINEERING
(AUTONOMOUS)
(Approved by AICTE, New Delhi and Affiliated to Andhra University)
VISAKHAPATNAM-530048

Institute Vision

To evolve into and sustain as a Centre of Excellence in Technological Education and Research with a holistic approach.

Institute Mission

- ***To produce high quality engineering graduates with the requisite theoretical and practical knowledge and social awareness to be able to contribute effectively to the progress of the society through their chosen field of endeavor.***
- ***To undertake Research & Development, and extension activities in the fields of Science and Engineering in areas of relevance for immediate application as well as for strengthening or establishing fundamental knowledge.***

Department Vision

To become a sought after center for higher learning and application in the field of Mechanical Engineering

Department Mission

- ***To produce competent and responsible mechanical engineering graduates and post graduates by imparting quality and value based education***
- ***To prepare students for professional career and guide them for entrepreneurship and higher studies including research***
- ***To motivate the young minds towards services beneficial to the society through their academic and professional activities***

Program Outcomes (POs): A graduate of mechanical engineering will be able to

PO-1	Apply the knowledge of mathematics, science, engineering fundamentals to solve complex mechanical engineering problems.
PO-2	Attain the capability to identify, formulate and analyse problems related to mechanical engineering.
PO-3	Design solutions for mechanical system components and processes that meet the specified needs with appropriate consideration for public health and safety.
PO-4	Perform analysis, conduct experiments and interpret data by using research methods such as design of experiments to synthesize the information and to provide valid conclusions..
PO-5	Select and apply appropriate techniques from the available resources and current mechanical engineering and software tools.
PO-6	Carry out their professional practice in mechanical engineering by appropriately considering and weighing the issues related to society.
PO-7	Understand the impact of the professional engineering solutions on environmental safety and legal issues.
PO-8	Transform into responsible citizens by resorting to professional ethics and norms of the engineering practice.
PO-9	Function effectively in individual capacity as well as a member in diverse teams and in multidisciplinary streams.
PO-10	Communicate fluently with the engineering community and society, and will be able to prepare reports and make presentations effectively.
PO-11	Apply knowledge of the engineering and management principles to managing projects and finance in multidisciplinary environments.
PO-12	Engage themselves in independent and life-long learning to continuing professional practice in their specialized areas of mechanical engineering.

Program Specific Objectives (PSOs): The student must attain the knowledge and skills to

PSO-1	Design, analyse and optimize mechanical systems along with control mechanisms
PSO-2	Manufacture mechanical components by selecting effective processing methods and efficient tools
PSO-3	Design, analyse and evaluate thermal systems

INTERNET OF THINGS LAB

Course code: 20CS11S3

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Course Outcomes: At the end of the course the student will be able to

CO1:	Build basic prototypes using Arduino Nano.
CO2:	Use different types of sensors, actuators for Arduino Nano.
CO3:	Demonstrate the setup and Installation procedure of Raspberry Pi.
CO4:	Build prototypes using Raspberry pi with different communication protocols.
CO5:	Design an interface using Tkinter to control the IoT devices.

LIST OF EXPERIMENTS

1. Install IDE of Arduino and write a program using Arduino IDE to blink LED.

2. Interface LED and buzzer with Arduino to buzz for a period of time.

3. Interface RGB LED with Aurdino to obtain different colours and brightness using PWM.

4. a) Control a servo motor using Arduino with an input given through a push button (e.g: When the push button is pressed the servo motor has to rotate by 15 degrees).

b) Rotate Stepper motor either clockwise or anti clockwise at 'n' number of steps using Arduino. 5. Write a program to read the data from the RFID tag and display the information on the display board using Arduino and control LED (e.g: if it is a valid card then the LED should be ON otherwise OFF).

6. Control any two actuators connected to the Arduino using Bluetooth/Wifi.

7. Interface analog/digital sensors with Arduino and analyse the corresponding readings. (Sensors like temperature, alcohol, humidity, pressure, gas, sound pollution, level, weight, flow, proximity, LDR, PIR, pulse, vibration, sound etc..)

8. Demonstration of setup & working of Raspberry Pi. (Students have to prepare the report for the same).

9. Interface RGB LED with Raspberry Pi to obtain different colours and brightness using PWM.

10. a) Interface an ultrasonic sensor with Raspberry pi to print distance readings on the monitor when the sensor changes its position.

b) Reading the data from an analog sensor with Raspberry using Arduino serial port or ADC MCP3208 using SPI.

11. Post/read the data to/from the cloud via MQTT broker with a Raspberry Pi.
12. Send real-time sensor data to a smartphone using Raspberry Pi onboard Bluetooth.
13. Interface Picamera module using Raspberry Pi to perform operations of PiCamera-API or OpenCV library.
14. Implement an intruder alert system that alerts through email
15. Implement remote monitoring of smoke alarm systems using Raspberry Pi.
16. Create a user interface using Tkinter to control the API's in Raspberry Pi.

Reference Books:

1.Donald Norris, Internet of things_ do-it-yourself projects with Arduino, Raspberry Pi, and Beagle Bone Black,1st Edition, McGraw-Hill,2015.

2. Adeal Javed Lake Zurich, Illinois, Building Arduino Projects for the Internet: Experiments with RealWorld Applications, 1st Edition,USA, A press,2016.

3. Yashavant Kanetkar, Shrirang Korde, 21 IOT Experiments, 1st Edition, BPB Publications, 2018.

4. Dr. Rajesh Singh, Dr. Anita Gehlot, Dr. Lovi Raj Gupta, Navjot Rathour, Mahendra Swain, Bhupendra Singh, IoT based Projects Realization with Raspberry Pi, NodeMCU and Arduino, 1st Edition, BPB Publications, 2020.

WEB REFERENCES:

1. https://www.arduino.cc/reference/en
2. https://create.arduino.cc/projecthub
3. https://maker.pro/raspberry-pi/tutorial
4. https://projects.raspberrypi.org/en/projects

Course Outcome versus **Program Outcomes:**

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO-1	2			2	3				1		-	-
CO-2	2			3	3				1		-	-
CO-3	2			2	3				1		-	-
CO-4	2			3	3				1		-	-
CO-5	2			3	3				1		-	-

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), put -: No Correlation

Course Outcome Versus Program Specific Outcomes:

COs	PSO1	PSO2	PSO3
CO-1	2		
CO-2	2		
CO-3	2		
CO-4	2		
CO-5	2		

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), put -: No Correlation

Assessment Methods:	Observation / Record / Internal Exam / End Exam

Teaching-Learning and Evaluation

Week	**Contents**	**Course Outcomes**	**Sample Questions**	**Teaching Learning Strategy**	**Assessment Method & Schedule**
1	Week-1	CO-1	Introduction to Arduno Installation procedure of Arduino IDE	Lecture/ Program Execution	End Exam
2.	Week-2	CO-1	Write a sample program for blinking single and multiple LEDS connected to Arduino	Lecture/ Program Execution	End Exam
3.	Week-3	CO-2	1. Develop a program to simulate the Push buttons with interfacing with A rduino	Lecture/ Program Execution	End Exam
4.	Week-4	CO-2	1. Write a Program to interface the Buzzerand LED with Arduino	Lecture/ Program Execution	End Exam
5.	Week-5	CO-2	1. Implement a program for PWMusing RGB LED 2. Show the different colors in RGB LED which is	Lecture/ Program Execution	End Exam

			already interfaced with Arduino		
6.	Week-6	CO-2	1.Implement a program to interface the servo motor with Aurdino 2.Write a program to interface thestepper motor with Aurdino	Lecture/ Program Execution	End Exam
7.	Week-7	CO-2	1. Implement a program to interface the RFID Sensor	Lecture/ Program Execution	End Exam
8.	Week-8	CO-2	1.Write a program to interface any actuators using Bluetooth or WiFi communication	Lecture/ Program Execution	End Exam
9.	Week-9	CO-2	1.Write a program to interface any digital sensors Implement a program to interface the analog sensors.	Lecture/ Program Execution	End Exam
10.	Week-10	CO-3	1.Demonstration of setup and Working procedure of Raspberry Pi	Lecture/ Program Execution	End Exam
11.	Week-11	CO-4	1.Write a program for implementation of PWM effect on RGB LED that is connected to Raspberry Pi	Lecture/ Program Execution	End Exam
12.	Week-12	CO-4	1.Implement a program to interface theUltrasonic sensor 2.Write a program to interface the Analog Sensors to the Raspberry Pi	Lecture/ Program Execution	End Exam
13.	Week-13	CO-4	1. Get/Post the data from/to the cloud 2.Send the sensor data to the smartphone using Bluetooth technology	Lecture/ Program Execution	End Exam
14.	Week-14	CO-4	1.Implement a program to interface the Pi-camera to the Raspberry Pi for capturing the images. 2.Develop a program to send a mail through Raspberry Pi device	Lecture/ Program Execution	End Exam

15.	Week-15	CO-5	**1.** Implement the Smoke alarm system that is connected remotely	Lecture/ Program Execution	End Exam
16	Week-16	CO-5	1. Create an user interface using Tkinter to control the API's of Raspberry Pi	Lecture/ Program Execution	End Exam
17	Week-17	Backlog Lab		Lecture/ Program Execution	End Exam
18	Week-18	Backlog Lab		Lecture/ Program Execution	End Exam
Final Examination					

Week-1

Install IDE of Arduino and write a program using Arduino IDE to blink LED.

Theory:

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them
Writing Sketches:Programs written using Arduino Software (IDE) are called **sketches**. These sketches are written in the text editor and are saved with the file extension .ino. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom righthand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

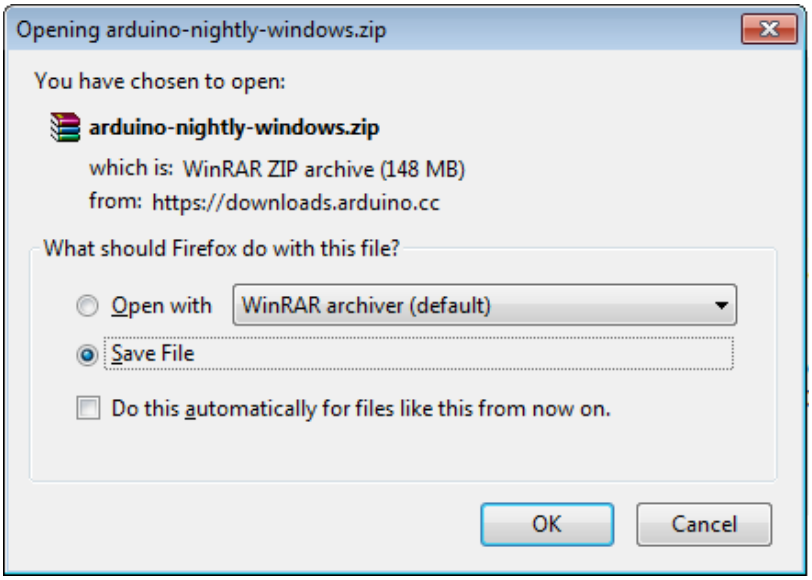
Aurdino Installation Steps:

Step-1:

choose an Arduino board and a USB cable. In case of Arduino UNO, Arduino Duemilanove, Nano, Arduino Mega 2560, or Diecimila, a standard USB cable (A plug to B plug is required.

Step 2 − Download Arduino IDE Software.

Different versions of Arduino IDE can be downloaded from the Download page on the Arduino Official website. Select software, which is compatible with the operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.



Step 3 − Power up Aurdino board.

The Arduino Uno, Mega, Duemilanove and Arduino Nano automatically draw power from either, the USB connection to the computer or an external power supplyConnect the Arduino board to your computer using the USB cable. The green power LED (labeled PWR) should glow.

Step 4 − Launch Arduino IDE.

After Arduino IDE software is downloaded, unzip the folder. Inside the folder, one can find the application icon with an infinity label (application.exe). Double-click the icon to start the IDE.

Once the software starts, you have two options −

- Create a new project.
- Open an existing project example.

To create a new project, select File → New.

Sample Viva Questions:

1. What is Arduino?
2. How many types of Arduino do we have?
3. What language is a typical Arduino code based on?
4. What language is the Arduino IDE built on?
5. How many analog pins are used in Arduino Mega board?
6. Arduino Codes are referred to as _____ in the Arduino IDE.

Week-2

2.a. AIM: To implement a program to blink LED every second.

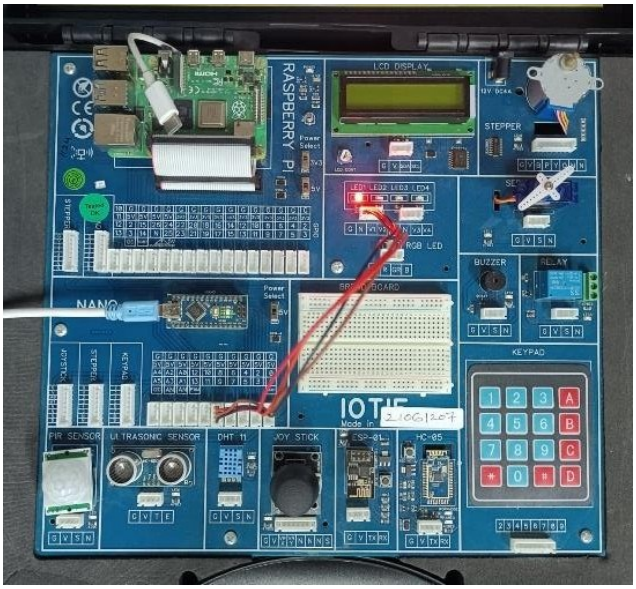
COMPONENTS REQUIRED: Arduino board**,** LED's**,** Resistors, Breadboard

Theory:

LED (Light Emitting Diode) is a semiconductor light emitting diode. We know that diode allows the current in one direction and does not allow the reverse current which will affect the components in the circuit. LED also do the same function but will emit a small light when it allowed the current, which will give the sign or visual indication to the normal human that circuit is working. There are lots of applications using LEDs. They are mainly used for visual indication in any electronic devices, measuring and interacting with the process, displaying the pictures in TV or in any advertisement hoarding, etc.

PROGRAM:

```
void setup(){
   pinMode(8,OUTPUT);
}
void loop(){
   digitalWrite(8,1);
   delay(1000);
   digitalWrite(8,0);
   delay(1000);
}
```



OUTPUT:

After starting the simulation, it sets the LED HIGH (Fig.2) i.e. turns ON the LED for 1000 milli seconds and after that it againsets the LED LOW (Fig.1) i.e turns OFF the LED for 1000 milli seconds. This process continues till we stop the simulation.

2.b.AIM: To implement a program to blink 2 LEDs every second alternatively.

COMPONENTS REQUIRED: Arduino board**,** LED's**,** Resistors, Breadboard

PROGRAM:

```
void setup(){
pinMode(8,OUTPUT;
pinMode(9,OUTPUT);
  }
  void loop(){
  digitalWrite(8,1);
```

```
digitalWrite(9,0);
delay(1000);
digitalWrite(8,0);
digitalWrite(9,1);
delay(1000);
}
```

OUTPUT :After starting the simulation, It sets the LED1 (right LED) to HIGH i.e, turns ON for 1000 milli seconds and then sets the LED1 (right LED) to LOW i.e, turns OFF for 1000 milli seconds. Now it sets the LED2 (left LED) to HIGH i.e, turns ON for 1000 milli seconds and then sets the LED2 (left LED) to LOW i.e, turns OFF for 1000 milli seconds. This process continues till we stop the simulation.

2.c.**AIM:** To implement a program to change the intensity of LED.

PROGRAM:

```
void setup()
{
  pinMode(8,OUTPUT);
}

void loop()
{
  for(int i=0;i<=255;i++)
  {
    analogWrite(8,i);
    delay(100);
  }
  for(int i=255;i>0;i--)
  {
    analogWrite(8,i);
    delay(100);
  }
}
```

Output:
After starting the simulation, It sets the intensity of LED1 (right LED) from 0 to 255. Intensity of 0 indicates LOW position and an intensity of 255 indicates HIGH. Once the intensity reaches 255, the intensity starts decreasing from 255(HIGH) to 0(LOW) .

Sample Viva Questions:

1. What is the use of the Arduino.h header file?
2. What does delay(5000) does?
3. digitalWrite(13, LOW); turns the light
4. LED is a _____ sensor.
5. Why are comments added in the Aurdino Code?

Week-3

Aim: Develop a program to simulate the Push buttons with interfacing with Arduino.

COMPONENTS REQUIRED: Arduino board**,** LED's **,** Resistors**,** Breadboard**,** Push button

Software Used: Tinker CAD

THEORY: We set the Button1 variable as integer 2 and we connect the button at pin 2 on the Board. Then the LED is connected to pin 3 using the resistor in series with it. In this code we can turn on the LED when we press the button then turn it off when we press the button again**.**

CODE:

```
int LED=3;
int Button1=2;
void setup()
{

pinMode(Button1, INPUT);
digitalWrite(Button1, HIGH);
pinMode(LED, OUTPUT);

}
void loop()
{
if(digitalRead(Button1)==LOW)
 {
digitalWrite(LED, HIGH);
 }
else
 {
digitalWrite(LED, LOW);

 }
}
```

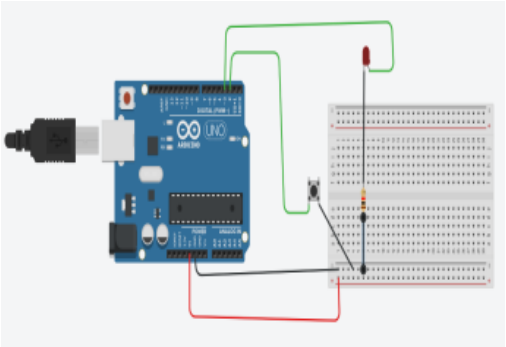


Fig.1: Circuit before starting the simulation

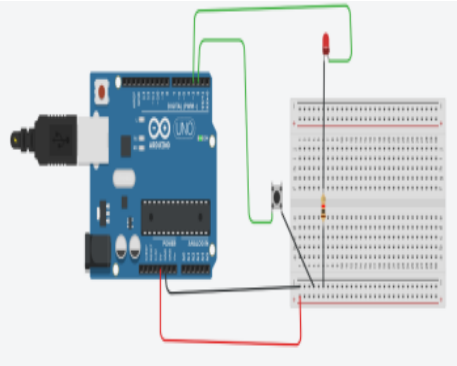


Fig.2: After starting the simulation

OUTPUT: After starting the simulation, when the push button is pressed the LED will be ON whereas if the pushbutton is not pressed the LED will be OFF.

2. AIM: To control two LED's with two push buttons

COMPONENTS REQUIRED: Arduino board, LED's **,** Resistors**,** Breadboard**,** Push buttons

THEORY: We set the Button1 variable as integer 2 and we connect the button at pin 2 on the Board. We set the Button2 variable as integer 4 and we connect the button at pin 4 on the Board. Then the LED is connected to pin 3 using the resistor in series with it and the LED2 is connected to pin 5 using the resistor in series with it. In this code we can turn on the LED's when we press the buttons then turn it off when we press the buttons again**.**

CODE:

```
int LED=3;
int Button1=2;
int LED2=5;
int Button2=4;
void setup()
{
pinMode(Button1, INPUT);
digitalWrite(Button1, HIGH);
pinMode(LED, OUTPUT);
pinMode(Button2, INPUT);
simulation digitalWrite(Button2, HIGH);

pinMode(LED2, OUTPUT);

}
void loop()
{
if(digitalRead(Button1)==LOW)
 {
digitalWrite(LED, HIGH);
 }
 else
 {
digitalWrite(LED, LOW);
 }
if(digitalRead(Button2)==LOW)
 {
digitalWrite(LED2, HIGH);
 }
 else
 {
digitalWrite(LED2, LOW);
 }
}
```

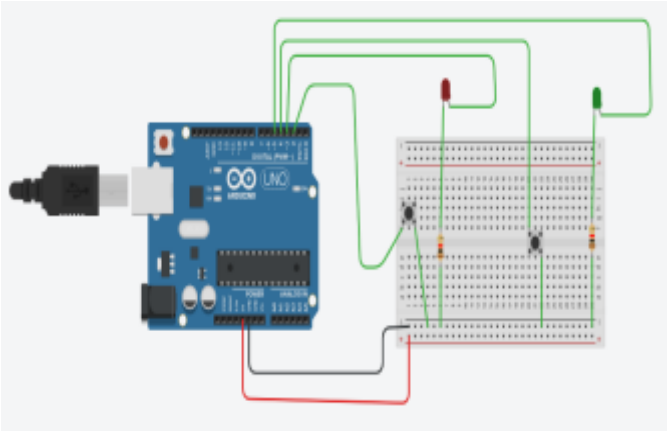
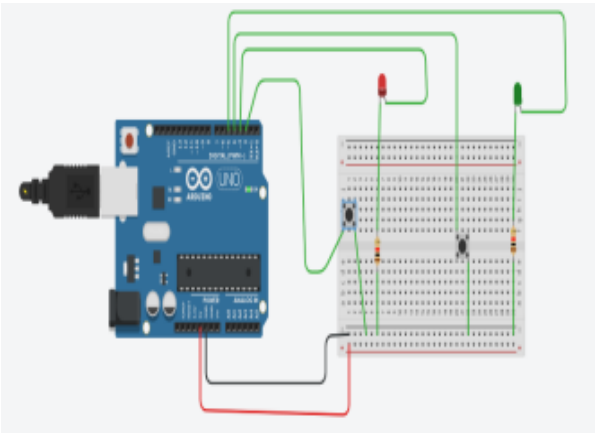


Fig.1: Circuit before starting the



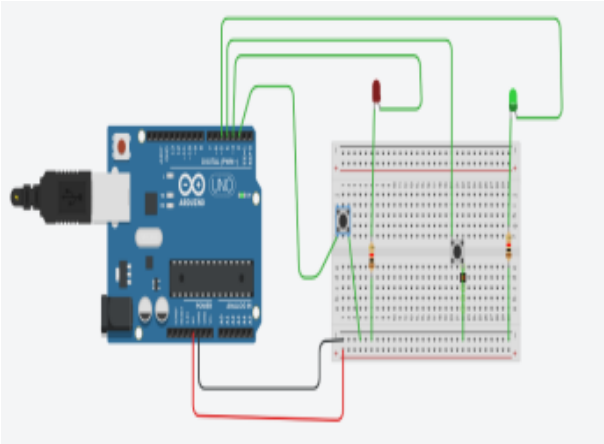


Fig.2: After starting the simulation, if the push button corresponding to the red LED is pressed then the Red LED will be ON and when the button is released Red LED will be OFF.

Fig.3: If the push button corresponding to the Green LED is pressed then the Green LED will be ON and when the button is released Green LED will be OFF.

OUTPUT:

After starting the simulation, if the push button corresponding to the red LED is pressed then the Red LED will be ON and when the button is released, Red LED will be OFF. After that, If the push button corresponding to the Green LED is pressed then the Green LED will be ON and when the button is released then the Green LED will be OFF.

Sample Viva Questions:

1. **What is th function of Resistors?**
2. **Give use cases where the LED-push button can be used?**
3. **What does G, V, N stand for?**
4. **Explain serial and parallel connections on PCB.**
5. **What is the difference between Aurdino Uno and Aurdino Nano?**
6. **How does push Buttons work?**

Week-4

Interface LED and buzzer with Arduino to buzz for a period of time.

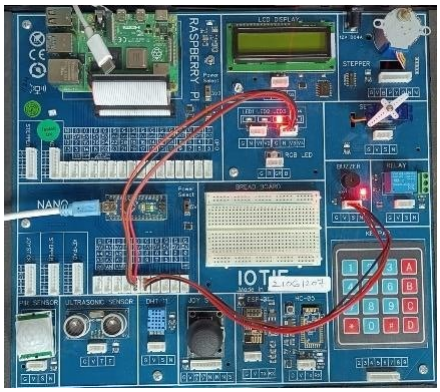
AIM: To implement a program to switch LED and buzzer ON every second.

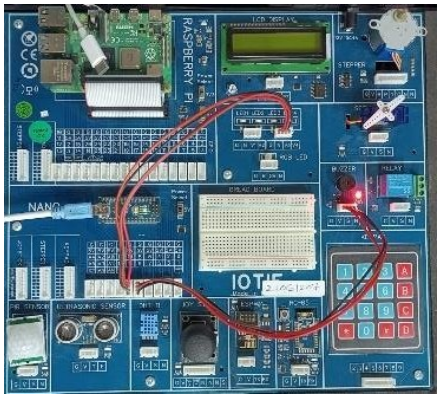
Theory:

It is a simple device which can generate beeps and tones. Working principle of the device is piezoelectric effect. The main component of this device is a piezo crystal, A special material that change shape when a voltage applied to it. In this code, if the push button corresponding to the LED is pressed then the LED will be ON and the buzzer starts

PROGRAM:

```
void setup(){
    pinMode(8,OUTPUT);
    pinMode(9,OUTPUT);
}
void loop(){
    digitalWrite(8,1);
    digitalWrite(9,1);
    delay(1000);
    digitalWrite(8,0);
    digitalWrite(9,0);
    delay(1000);
}
```





Output:

After starting the simulation, if the push button corresponding to the red LED is pressed then the Red LED will be ON and the buzzer starts buzzing. When the button is released then the LED will be OFF and the buzzer stops buzzing.

Sample Viva Questions:

1. Give examples of cases where LED and Buzzer are interfaced with Aurdino?
2. Which is the software or a programming language used for controlling of Arduino
3. What would happen if we used the following code but the wire was not plugged into pin 13?
   digitalWrite(13, HIGH);
   delay(1000);
   digitalWrite(13, LOW);
   delay(1000);
4. Arduino IDE consists of 2 functions. What are they?
5. A buzzer is an _____ component.

WEEK-5

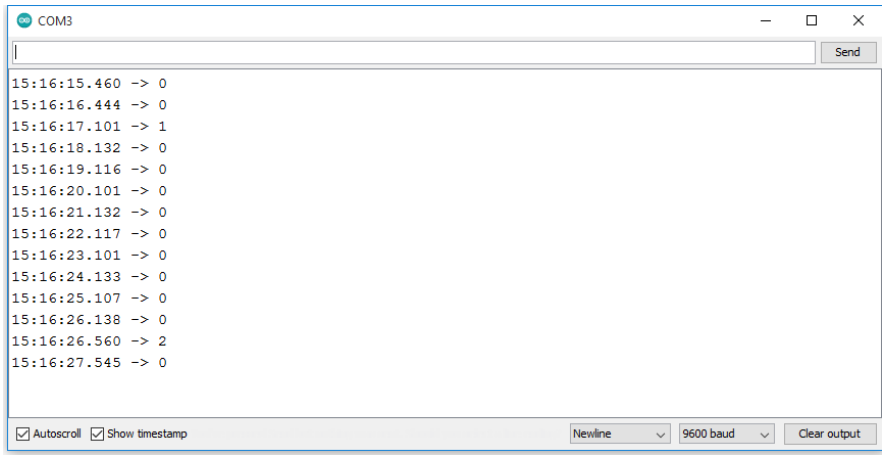
Interface RGB LED with Arduino to obtain different colours and brightness using PWM

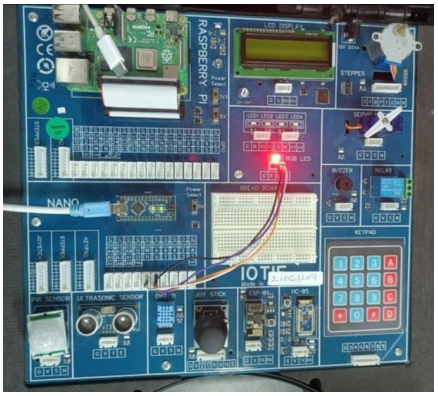
Theory: With an RGB LED you can, of course, produce red, green, and blue light, and by configuring the intensity of each LED, you can produce other colors as well.For example, to produce purely blue light, you'd set the blue LED to the highest intensity and the green and red LEDs to the lowest intensity. For a white light, you'd set all three LEDs to the highest intensity. PWM is a method of changing the perceived and actual brightness of an LED by pulsing the power to the LED on and off. This is done very rapidly so there should be no perceptible flicker, but changing the average power will change the brightness.To produce other colors, you can combine the three colors in different intensities. To adjust the intensity of each LED you can use a PWM signal.

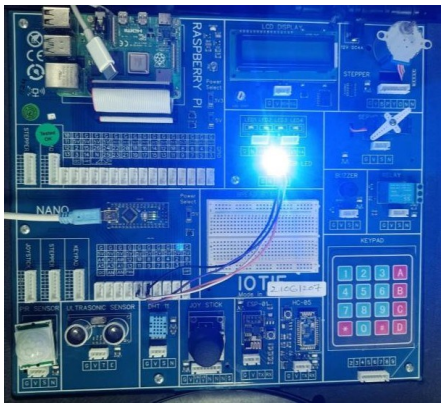
5.1 AIM: To implement a program to obtain red, blue and green colors of RGB LED with userinput.
COMPONENTS REQUIRED: Arduino board , LED, connecting wires, resistor

PROGRAM:

```
int c;
void setup() {
  Serial.begin(9600);
  pinMode(10,OUTPUT);
  pinMode(8,OUTPUT);
  pinMode(9,OUTPUT);
}
void loop()
{
  c = Serial.parseInt();
  if(c==1) {
    digitalWrite(8,1);
    digitalWrite(9,0);
    digitalWrite(10,0);
  }
  else if(c==2) {
    digitalWrite(9,1);
    digitalWrite(8,0);
    digitalWrite(10,0);
  }
  else if(c==3) {
    digitalWrite(10,1);
    digitalWrite(8,0);
    digitalWrite(9,0);
  }
  Serial.println(c);
```







Output: The RGB LED displays the required color as per user input.

5.2 AIM: To implement a program to obtain different colors from RGB LED through user input.

COMPONENTS REQUIRED: Arduino board , LED, connecting wires, resistor

PROGRAM:

```
String color = "";
void setup()
{
  Serial.begin(9600);
  pinMode(9,OUTPUT);
  pinMode(10,OUTPUT);
  pinMode(11,OUTPUT);
}
void loop()
{
  color = Serial.readString();
  color.trim();

  if(color == "red" || color == "RED") {
    rgb(255,0,0);
  }
  if(color == "green" || color == "GREEN") {
    rgb(0,0,255);
  }
  if(color == "blue" || color == "BLUE") {
    rgb(0,255,0);
  }
  if(color == "yellow" || color == "YELLOW") {
    rgb(255,0,255);
  }
  if(color == "cyan" || color == "CYAN") {
    rgb(0,255,255);
  }
  if(color == "magenta" || color == "MAGENTA") {
    rgb(255,255,0);
  }
  if(color == "purple" || color == "PURPLE") {
    rgb(128,128,0);
  }
  if(color == "navyblue" || color == "NAVYBLUE") {
    rgb(0,128,0);
  }
  if(color == "brown" || color == "BROWN") {
    rgb(165,42,42);
  }
```

```
  if(color == "sage" || color == "SAGE") {
    rgb(178,136,172);
  }
}

void rgb(int x, int y, int z)
{
  analogWrite(9,x);
  analogWrite(10,y);
  analogWrite(11,z);
}
```

Output: The RGB LED displays the required color as per user input.

5.3 AIM: To implement a program to change the intensity of RGB LED.

COMPONENTS REQUIRED: Arduino board , LED, connecting wires, resistor

PROGRAM:

```
void setup()
{
  pinMode(9,OUTPUT);
  pinMode(10,OUTPUT);
  pinMode(11,OUTPUT);

}

void loop()
{

  for(int i = 0; i <=255; i++)
  {
   analogWrite(9,i);
   analogWrite(10,i);
   analogWrite(11,i);
   delay(100);
  }
  for(int i = 255; i >=0; i--)
  {
   analogWrite(9,i);
   analogWrite(10,i);
   analogWrite(11,i);
   delay(100);
```

```
  }

}
```

Output: The RGB LED intensity continuous displays a change in intensity .

5.4 AIM: To implement a program to increase the intensity of RGB LED.

COMPONENTS REQUIRED: Arduino board , LED, connecting wires, resistor

PROGRAM:

```
int i;
void setup() {}
void loop()
{
  for(int i=0;i<=255;i++)
  {
    RGB(0,i,0);
    delay(100);
  }
  for(int i=0;i<=255;i++)
  {
    RGB(i,0,0);
    delay(100);
  }
  for(int i=0;i<=255;i++)
  {
    RGB(0,0,i);
    delay(100);
  }
}

void RGB()
{
  analogWrite(9,R);
  analogWrite(10,G);
  analogWrite(11,B);
}
```

Output: The RGB LED intensity continuous displays a change in intensity in the order of red, blue, green.

Sample Viva Questions:

1. What is PWM?
2. What is color code for red?
3. What does the 4 pins for RGB Led indicate?
4. How do we obtain white color in RGB LED?
5. How is intensity changed in PWM method?
6. What is duty cycle?

WEEK-6

a) Control a servo motor using Arduino with an input given through a push button (e.g: Whenthe push button is pressed the servo motor has to rotate by 15 degrees). b) Rotate Stepper motor either clockwise or anti clockwise at 'n' number of steps using Arduino.

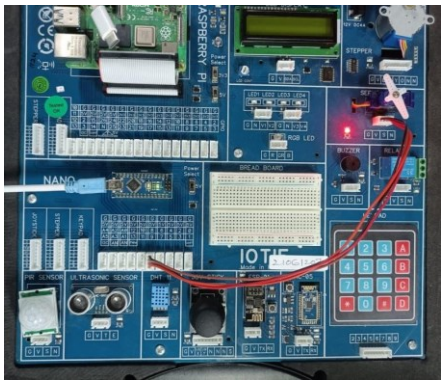
THEORY- Servomotor:

Servo motors are a class of motors where you can tell the motor, what position it should move to or what speed it should move .The name servo motor is related to the term servomechanism, which means that the motor is constantly monitored to control its motion. Servo motors move, stop, and start conveyor belts carrying product along to various stages, for example, in product packaging/bottling, and labelling.

1. **AIM:** To implement a program to rotate servo motor 90^0 for every 500ms.

COMPONENTS REQUIRED: arduino board, servo motor, connecting wires

PROGRAM:



```
#include<Servo.h>
Servo s1;
int servopin=8;
void setup(){
   s1.attach(servopin);
}
void loop(){
   s1.write(90);
   delay(500);
}
```

Output: The servomotor rotates for 90^0 , a delay of 500 ms and comes back to initial position.

2.AIM: To implement a program to rotate servo motor 180^0 in clockwise direction.

COMPONENTS REQUIRED: arduino board, servo motor, connecting wires

PROGRAM:

```
#include<Servo.h>
Servo s1;
int servopin=8;
void setup(){
   s1.attach(servopin);
}
void loop(){
```

```
    for(i=0;i<=180;i+=45){
    s1.write(i);
    delay(500);
    }
}
```

Output: The servomotor rotates for 180^0 , a delay of 500 ms and comes back to initial position.

3.AIM: To implement a program to rotating servo motor 180^0 in anti-clockwise direction.

COMPONENTS REQUIRED: arduino board, servo motor, connecting wires

PROGRAM:

```
#include<Servo.h>
Servo s1;
int servopin=8;
void setup(){
   s1.attach(servopin);
}
void loop(){
for(i=180;i>=0;i--){
   s1.write(i);
   delay(500);
   }
}
```

Output: The servomotor rotates for 180^0 , a delay of 500 ms and comes back to initial position.

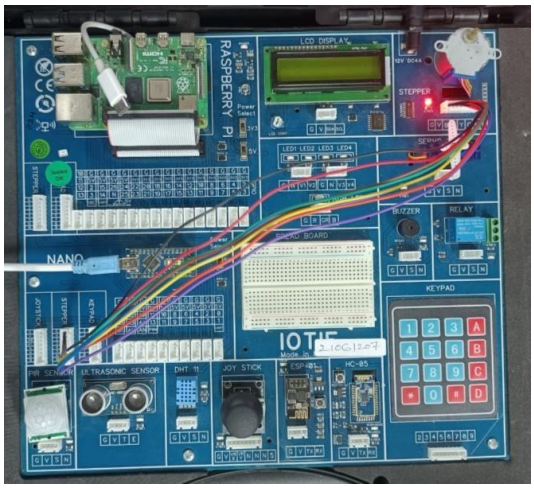
4.AIM: To implement a program to rotate stepper motor 360^0 in clockwise and then in anti-clockwise direction.

THEORY: It consist of 1 Stepper Motor of 200 steps 1.8° which is capable of rotate 200 steps per rotation this means that for each step it rotates approximately 1.8° (360/200).It will move over a certain number of steps in one direction or the other, and tell it the speed at which to step in that direction.

COMPONENTS REQUIRED: arduino board, stepper motor, connecting wires

PROGRAM:

```
#include<Stepper.h>     int
steps_per_rev=32;       int
gear_reduction=64;
```




```
int        steps_req=gear_reduction*steps_per_rev;
Stepper motor(steps_per_rev,2,4,3,5);
void                setup(){
  Serial.begin(9600);
}
void loop(){ Serial.println(steps_req);motor.setSpeed(900)
  motor.step(steps_req);
  delay(1000);
  Serial.println(steps_req);
  motor.setSpeed(900);
  motor.step(-steps_req);
  delay(1000);
}
```

OUTPUT: The motor will take one revolution in one direction, then one revolution in the other direction

5. AIM: To implement a program to rotate stepper motor 180^0 in clockwise and then 90^0 in anti-clockwise direction.

COMPONENTS REQUIRED: arduino board, stepper motor, connecting wires

PROGRAM:

```
#include<Stepper.h>     int
steps_per_rev=32;       int
gear_reduction=64;
int        steps_req=gear_reduction*steps_per_rev;
Stepper motor(steps_per_rev,2,4,3,5);
void                setup(){
  Serial.begin(9600);
}
void loop(){
  Serial.println(steps_req);
  motor.setSpeed(900);
  motor.step(steps_req/2);
  delay(1000);
  Serial.println(steps_req);
```

```
    motor.setSpeed(900);
    motor.step(-steps_req/4);
    delay(1000);
  }
```

OUTPUT: The motor will take one revolution in one direction, then one revolution in the other direction

Sample Viva Questions:

1. What is the difference between a stepper motor and servo motor?
2. What are the applications of Stepper motor?
3. What are the applications of servo motor?
4. How many terminals are there for servomotor?
5. How many pins are needed for stepper motor?

Week-7

To interface RFID/PIR sensor with Aurdino

AIM: To interface PIR sensor for detecting motion of an object.

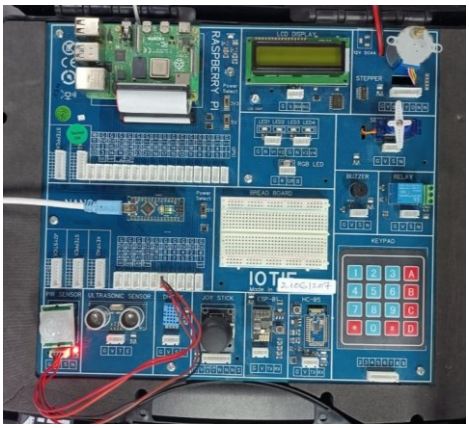
COMPONENTS REQUIRED: PIR sensor, connecting wires , Arduino board

THEORY:

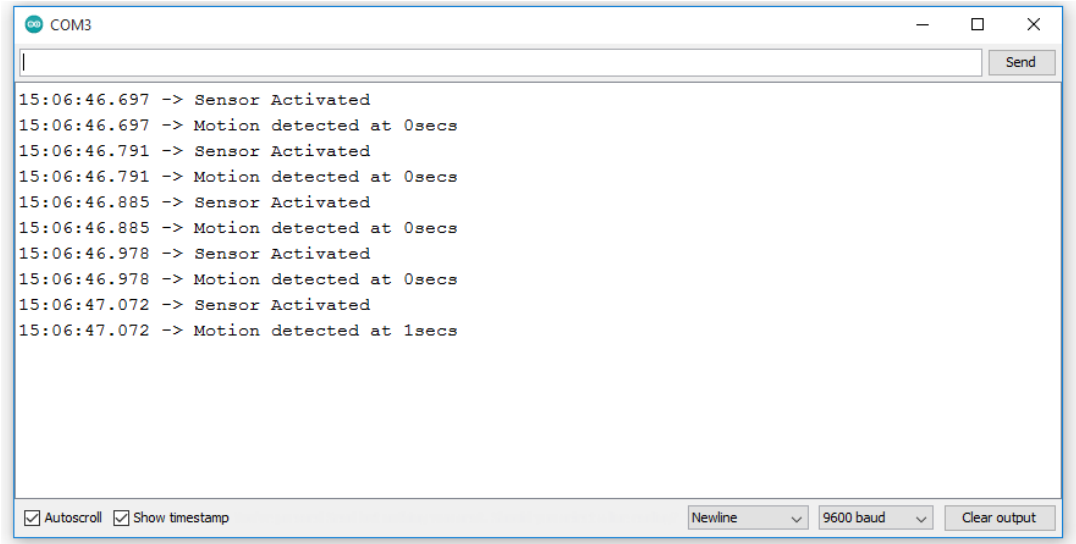
The PIR motion sensor is ideal to detect movement. PIR stand for "Passive Infrared". Basically, the PIR motion sensor measures infrared light from objects in its field of view. So, it can detect motion based on changes in infrared light in the environment. It is ideal to detect if a human has moved in or out of the sensor range.

PROGRAM:

```
int sensordata;void
setup(){
    Serial.begin(9600);
    pinMode(13,OUTPUT);
    pinMode(8,INPUT);
}
void loop(){ sesnsordata=digitalRead(8);
    if(sensordata==HIGH){
        digitalWrite(13,HIGH); Serial.println("Sensor
        Activated");Serial.print("Motion detected at");
        Serial.print(millis()/1000);
        Serial.println("secs");
        delay(50);
    }
    else{
        digitalWrite(13,LOW);
    }
    delay(50);
}
```



OUTPUT:



A message "Motion is detected at __secs" is displayed on serial Monitor.

Sample Viva Questions:

1. What is PIR Motion Sensor?
2. What is the function of PIR sensor?
3. What is the range of PIR sensor?
4. Is analog sensor or digital sensor?
5. Give applications of PIR sensor.

WEEK-8

Control any two actuators connected to the Arduino using Bluetooth/Wifi.

1.AIM: To implement a program to send and receive messages using Bluetooth module with Arduino.

COMPONENTS REQUIRED: Bluetooth HC-05 module, connecting wires ,Arduino board

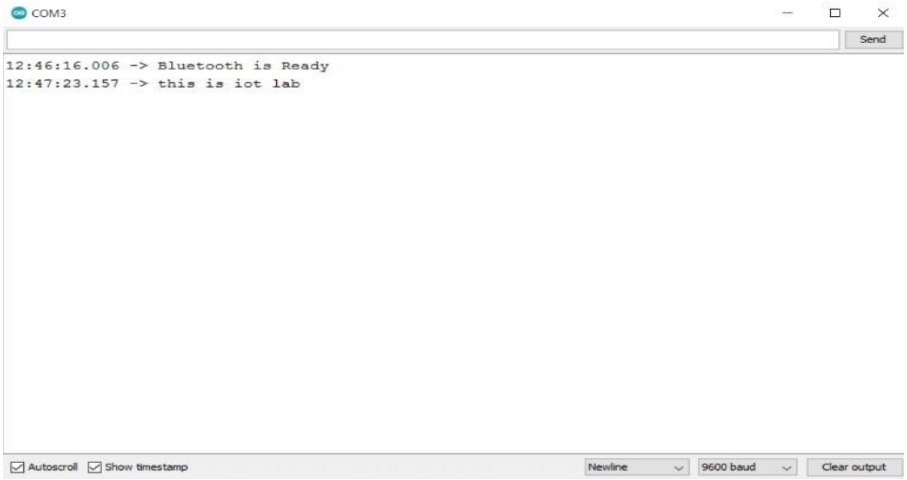
THEORY:

As Bluetooth Module HC-05 works on serial communication. It receives the data from the app and sends it through TX pin of Bluetooth module to RX pin of Arduino. The uploaded code inside Arduino checks the data received. The received data is displayed on Serial Monitor.

Program:

```
#include <SoftwareSerial.h>
SoftwareSerial EEBLUE(10,11);
void setup()
{
  Serial.begin(9600);
  EEBLUE.begin(9600);
  Serial.println("Bluetooth is ready ");
}
void loop()
{
  if(EEBLUE.available())
  Serial.write(EEBLUE.read());
  if(Serial.available())
  EEBLUE.write(Serial.read());
}
```

OUTPUT:

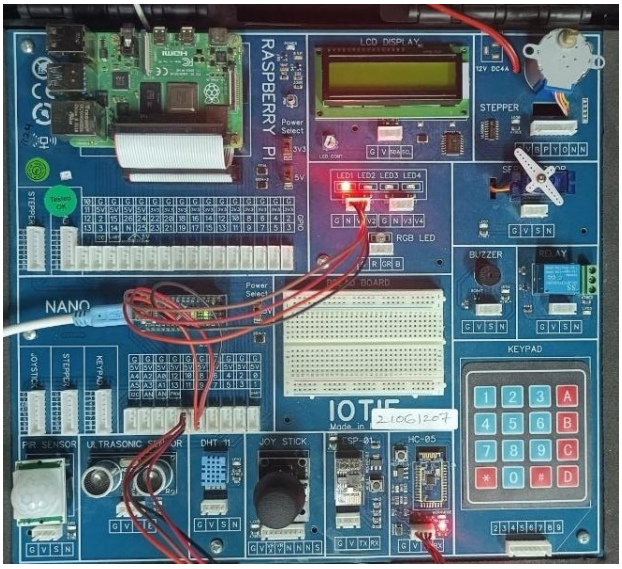


The Aurdino is interfaced with Bluetooth to send and receive data.

2.AIM: To implement a program using Bluetooth to interface LED based on Bluetoothterminal input.

COMPONENTS REQUIRED: Bluetooth HC-05 module, connecting wires ,Arduino board

PROGRAM:




```
#include <SoftwareSerial.h>
SoftwareSerial EEBLUE(10,11);int
input;
void setup()
{
   Serial.begin(9600);
   EEBLUE.begin(9600);
   Serial.println("BLUETOOTH IS READY");
   pinMode(8,OUTPUT);
}
void loop()
{
   if(EEBLUE.available()){
     input=EEBLUE.parseInt();
     if(input==1) digitalWrite(8,1);
     if(input==0) digitalWrite(8,0);


   }
}
```

Output: . If the receive data is 1, the LED turns ON, and if the received data is 0 the LED turns OFF.

3.AIM: To interface a servo motor using Bluetooth terminal.

COMPONENTS REQUIRED: Bluetooth HC-05 module, connecting wires ,Arduino board

THEORY: The HC-05 is an expensive module that is compatible with wide range of devices including smartphone, laptops and tablets. The HC-05 is a popular module and it communicates with the help of USART at 9600 Baud rate hence it is easy to interface with any microcontroller that supports USART. The android app sends data packets to the Bluetooth module. The Bluetooth modules send this data packet to Arduino Uno through Serial Communication. Arduino Uno is programmed to generate control signal for the servo motor depending upon the value of the data packet.

PROGRAM:

```
#include <SoftwareSerial.h>
#include<Servo.h> SoftwareSerial
EEBLUE(10,11);Servo s1;
```

```
int angle; void
setup()

{

   Serial.begin(9600);
   EEBLUE.begin(9600);

   Serial.println("BLUETOOTH IS READY");
   s1.attach(8);

}

void loop()

{

   if(EEBLUE.available()){
     angle=EEBLUE.parseInt(); for(int
     i=0;i<=angle;i+=20){s1.write(i);

       delay(500);

     }

   }

}
```

OUTPUT:

Whether the input can be given through the serial monitor window or by using the Bluetooth terminal application based on the value or number given by the user the servomotor starts to rotate for that particular angle of rotation only in both clockwise and anticlockwise direction one after the other and they are having a certain delay after each operation.

Sample Viva Questions:

1. Give examples of use cases where Bluetooth communication is used.
2. How does Bluetooth work?
3. What is the range of Bluetooth communication?
4. What are the different types of communication protocols?

WEEK-9

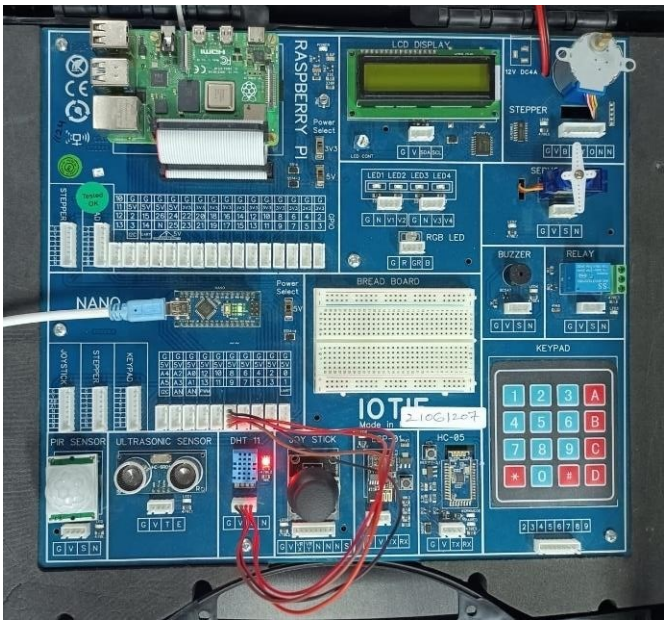
Interface analog/digital sensors with Arduino and analyse the corresponding readings.(Sensors like temperature, alcohol, humidity, pressure, gas, sound pollution, level, weight, flow, proximity, LDR, PIR, pulse, vibration, sound etc..)

1.AIM: To interface DHT11 sensor for measuring humidity and temperature.

COMPONENTS DHT sensor , connecting wires , arduino board **REQUIRED:**

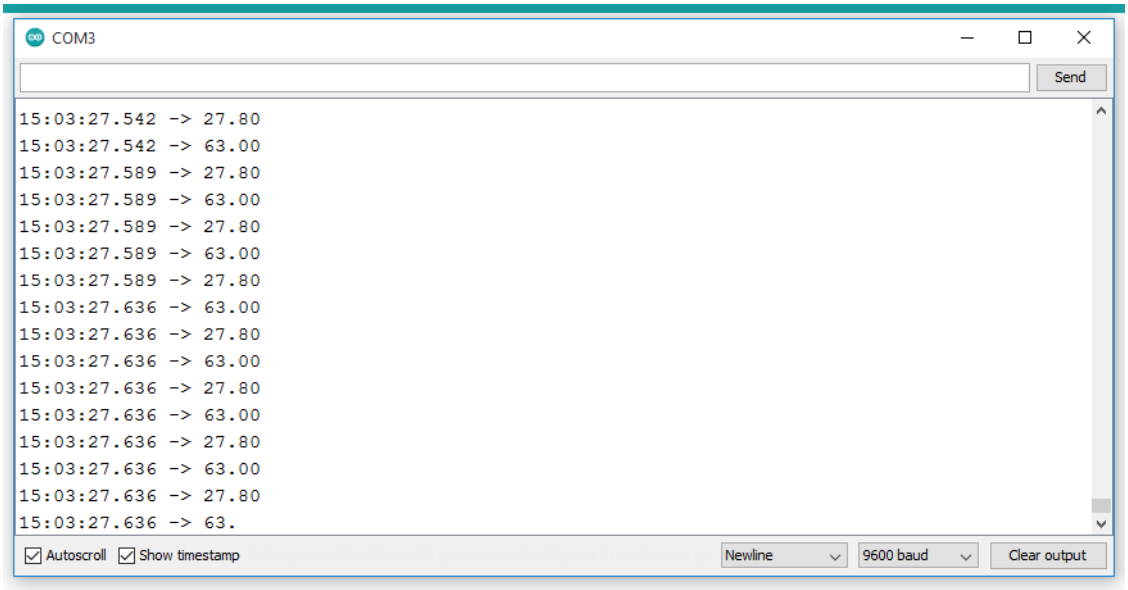
THEORY: The DHT11 is a commonly used Temperature and humidity sensor for prototypes monitoring the ambient temperature and humidity of a given area. The sensor can measure temperature from 0°C to 50°C with an accuracy of ±2°C and humidity from 20% to 90% with an accuracy of ±5% RH.

PROGRAM:




```
#include<DHT.h> DHT
dht(8,DHT11)
float t,h;
void setup(){
   Serial.begin(9600);
   dht.begin();
   Serial.println("Starting DHT Test");
   delay(2000);
}
void loop(){
   h=dht.readHumidity();
   t=dht.readTemperature();
   if(isnan(h)||isnan(t)){
      Serial.println("Failed to read");
   }
   else{
      Serial.println(h);
      Serial.println(t);
   }
}
```

Output:



DHT sensor placed in any environment detects the level of humidity and temperature values in that place and it is printed continuously in the serial monitor until the supply given to the sensor is halted

2.AIM: **To interface ultrasonic sensor for detecting the presence and measuring the distance of an object.**

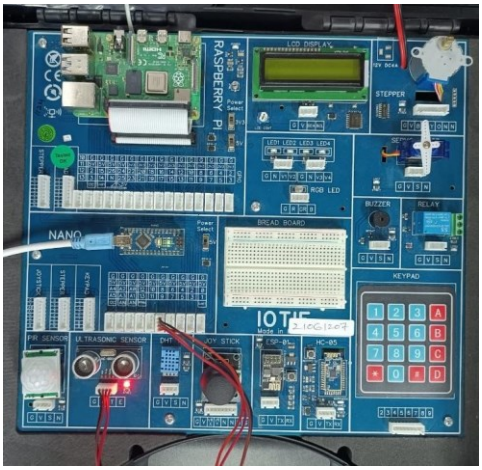
COMPONENTS :Ultrsaonic sensor , connecting wires , arduino board **REQUIRED:**

Theory:

An ultrasonic sensor is an instrument that measures the distance to an object using ultrasonic sound waves. An ultrasonic sensor uses a transducer to send and receive ultrasonic pulses that relay back information about an object's proximity. The working principle of this module is simple. It sends an ultrasonic pulse out at 40kHz which travels through the air and if there is an obstacle or object, it will bounce back to the sensor. By calculating the travel time and the speed of sound, the distance can be calculated.

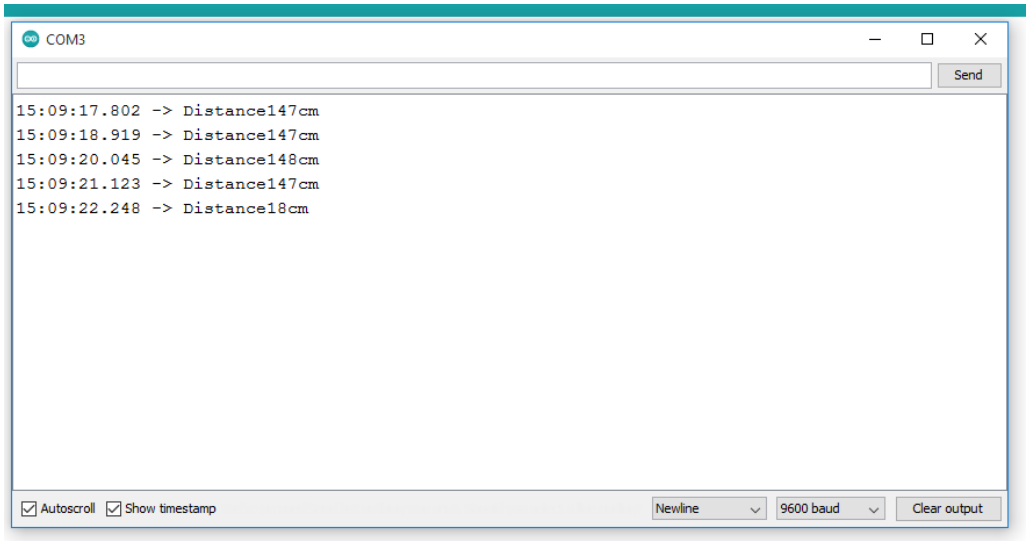
PROGRAM:

```
int distance,duration;void
setup(){
   Serial.begin(9600);
   pinMode(8,OUTPUT);
   pinMode(9,INPUT);
}
void loop(){ digitalWrite(8,HIGH);
   delay(100);
```



```
    digitalWrite(8,LOW);
    duration=pulseIn(9,1);
    distance=duration*0.032/2;
    Serial.print("Distance");
    Serial.print(distance);
    Serial.println("cm");
    delay(1000);
}
```

OUTPUT:



The distance is measured continously and is displayed in the serial window until interrupted by the keyboard.

3.AIM: To implement a program to use joystick with serial monitor.

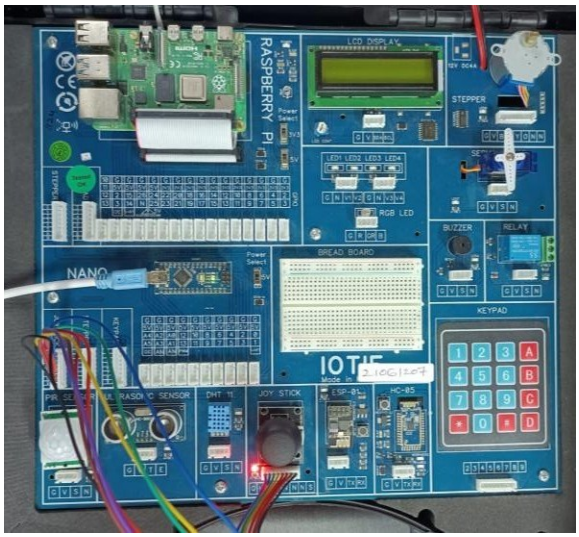
COMPONENTS REQUIRED: Arduino board, Ioystick, connecting probes

THEORY:

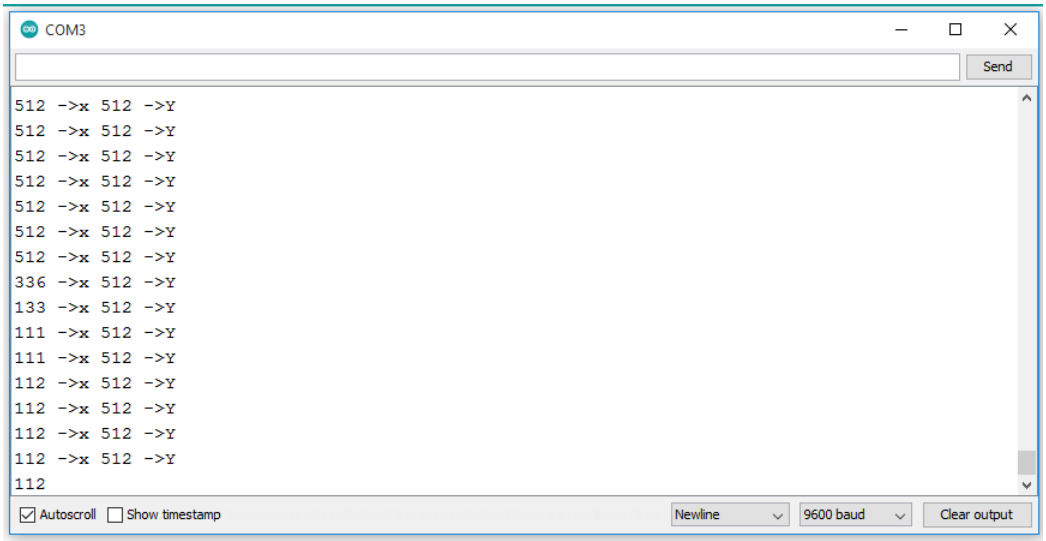
"**Joystick**" is simply a game controller. Because everybody has seen it while playing video games. But, apart from gaming, it can wide range of applications in electronics. Actually, the joystick is the combination of two potentiometers for the X and Y Plane. It gives analog data to the Arduino by reading the voltage from the potentiometer. This analog value get changes as we move the joystick shaft.

PROGRAM:

```
int VRx = A0; int
VRy  =  A1;  int
SW = 2;

int xPosition = 0; int
yPosition  =  0;  int
SW_state  =  0;  int
mapX = 0;

int mapY = 0;

void setup() {
  Serial.begin(9600);

  pinMode(VRx, INPUT);
  pinMode(VRy, INPUT);
  pinMode(SW, INPUT_PULLUP);

}

void loop() {

  xPosition   =   analogRead(VRx);
  yPosition   =   analogRead(VRy);
  SW_state = digitalRead(SW);

  mapX = map(xPosition, 0, 1023, -512, 512);

  mapY = map(yPosition, 0, 1023, -512, 512);


  Serial.print("X: ");
  Serial.print(mapX);
  Serial.print(" | Y: ");
  Serial.print(mapY);
  Serial.print(" | Button: ");
  Serial.println(SW_state);

  delay(100);

}
```



OUTPUT:



OUTPUT: Thus Joystick measured and displayed the X-axis,Y-axis and Z-axis direction.

Sample Viva Questions:

1. What is an IR sensor?
2. List different types of Analog and digital sensors.
3. What are the terminals of a joystick?
4. Explain the working of Ultrasonic sensor?
5. What is the range and resolution of DHT sensor?

Week-10

AIM: Demonstration of setup & working of Raspberry Pi

COMPONENTS REQUIRED: Raspberry pi

THEORY:

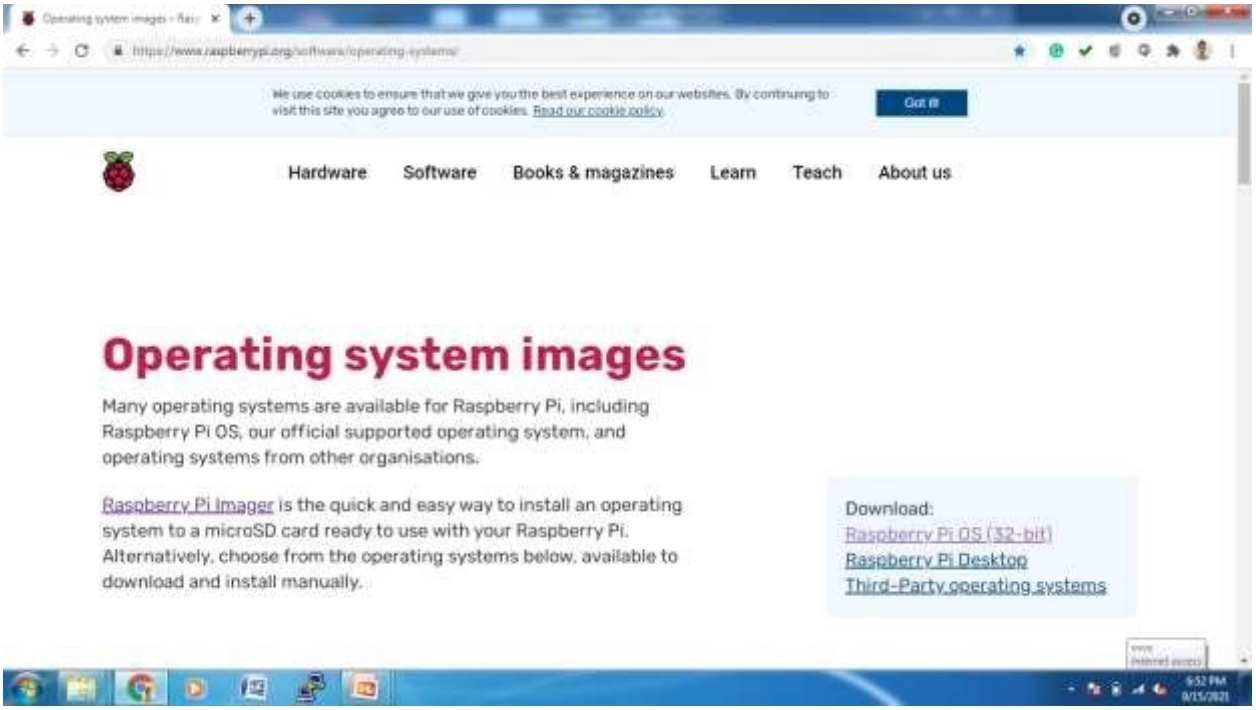
Raspberry Pi is a series of small single-board computers (SBCs) developed in the United Kingdom by the Raspberry Pi Foundation in association with Broadcom. The Raspberry Pi project originally leaned towards the promotion of teaching basic computer science in schools and in developing countries. The original model became more popular than anticipated, selling outside its target market for uses such as robotics. It is widely used in many areas, such as for weather monitoring, because of its low cost, modularity, and open design. It is typically used by computer and electronic hobbyists, due to its adoption of HDMI and USB devices.

Steps to install RaspberryPi

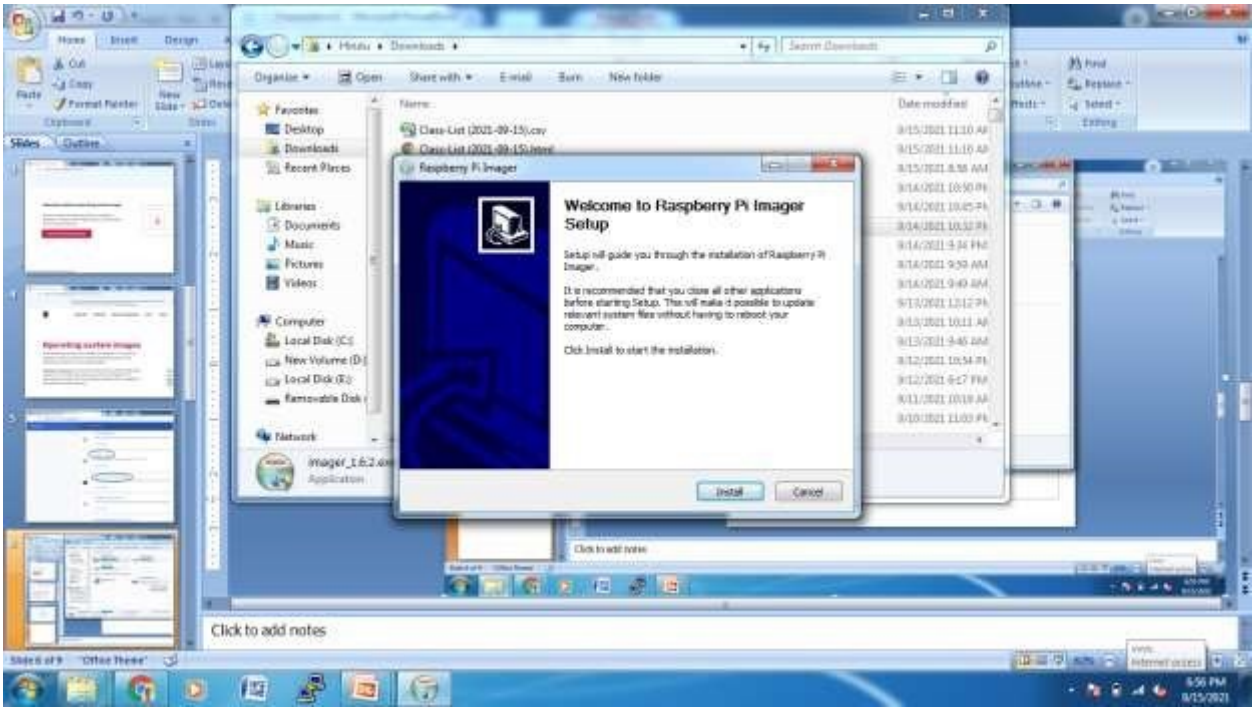
1) Go to google search and type pi imager windows and click on 'Raspberry Pi OS'

2) Inside the website click on 'Download for Windows'

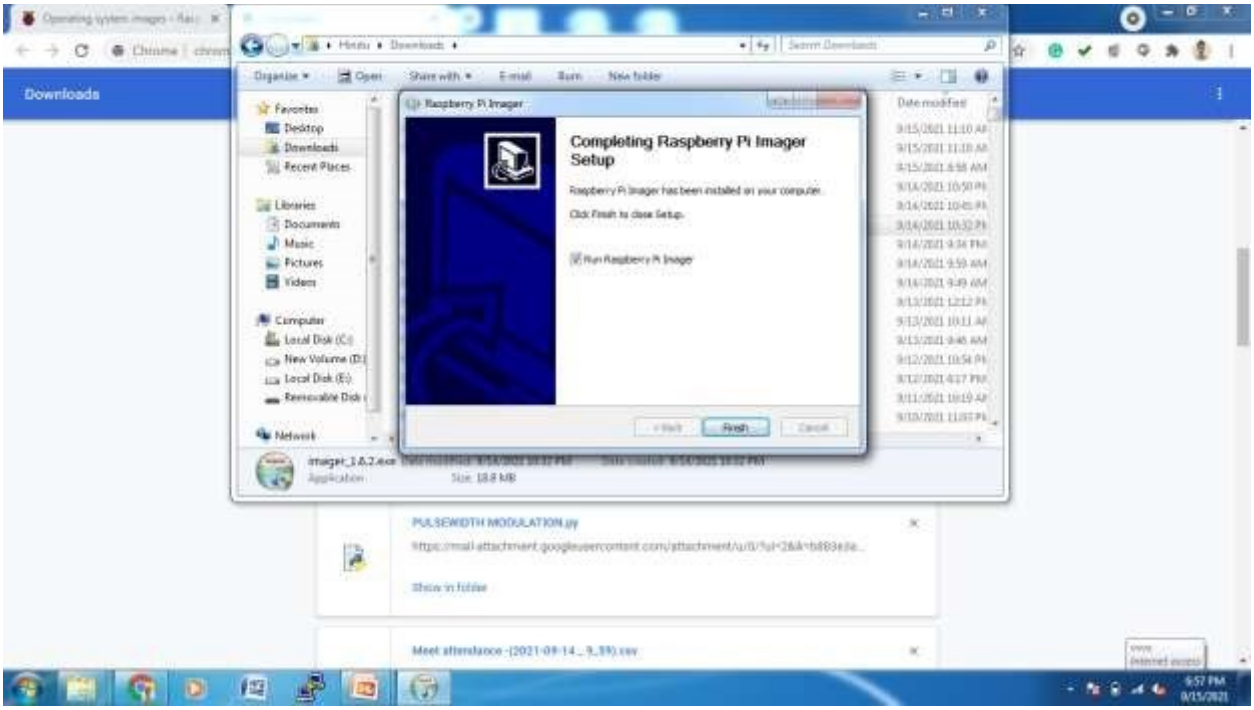
3) Next click on Raspberry Pi OS(32-bit)



4) The below window will pop up showing 'Welcome to RaspberryPi Imager Setup' and asks to install. Click on install



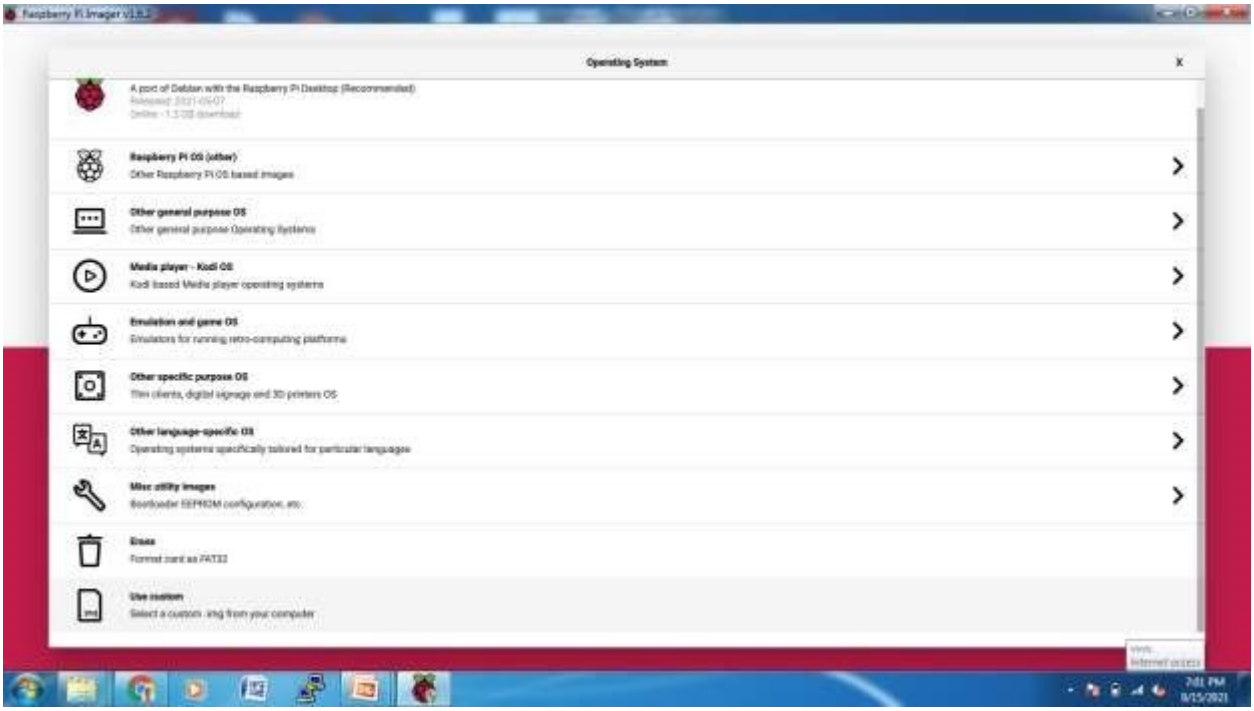
5) After installing,another window will apper as below.Click on finish to proceed further

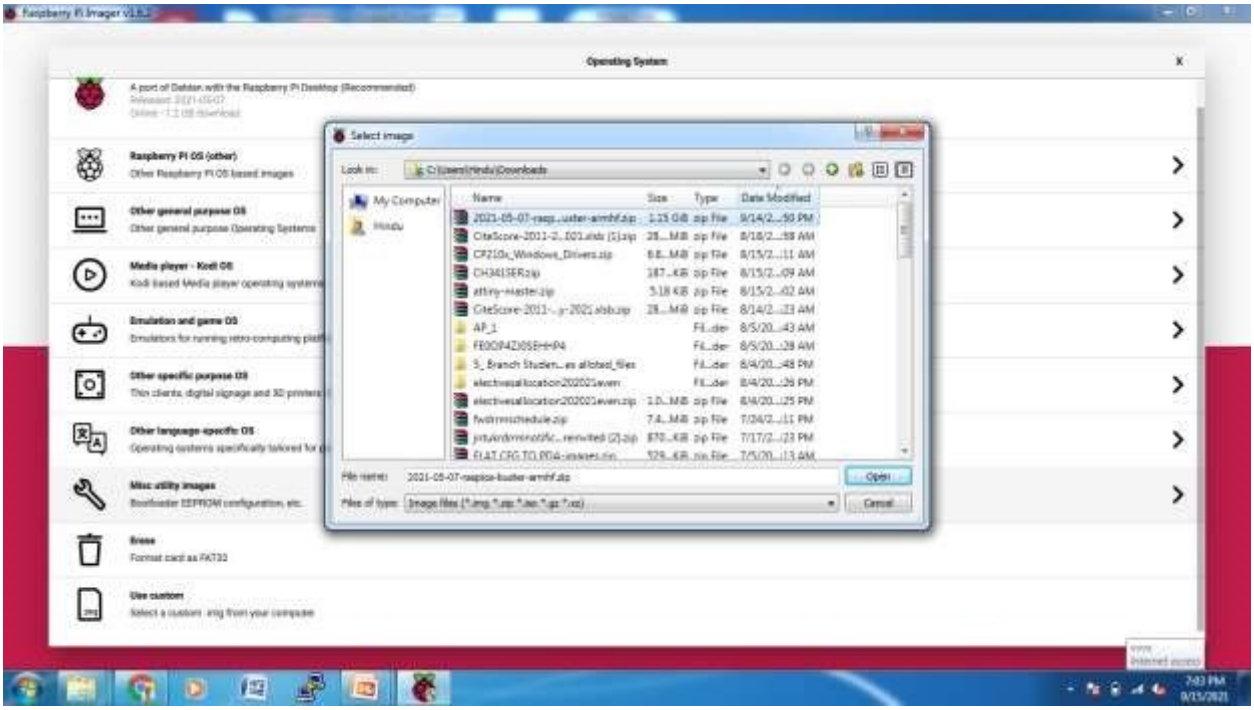


6) Raspberry Pi imager is installed as shown in the below figure.Now click on the the imager

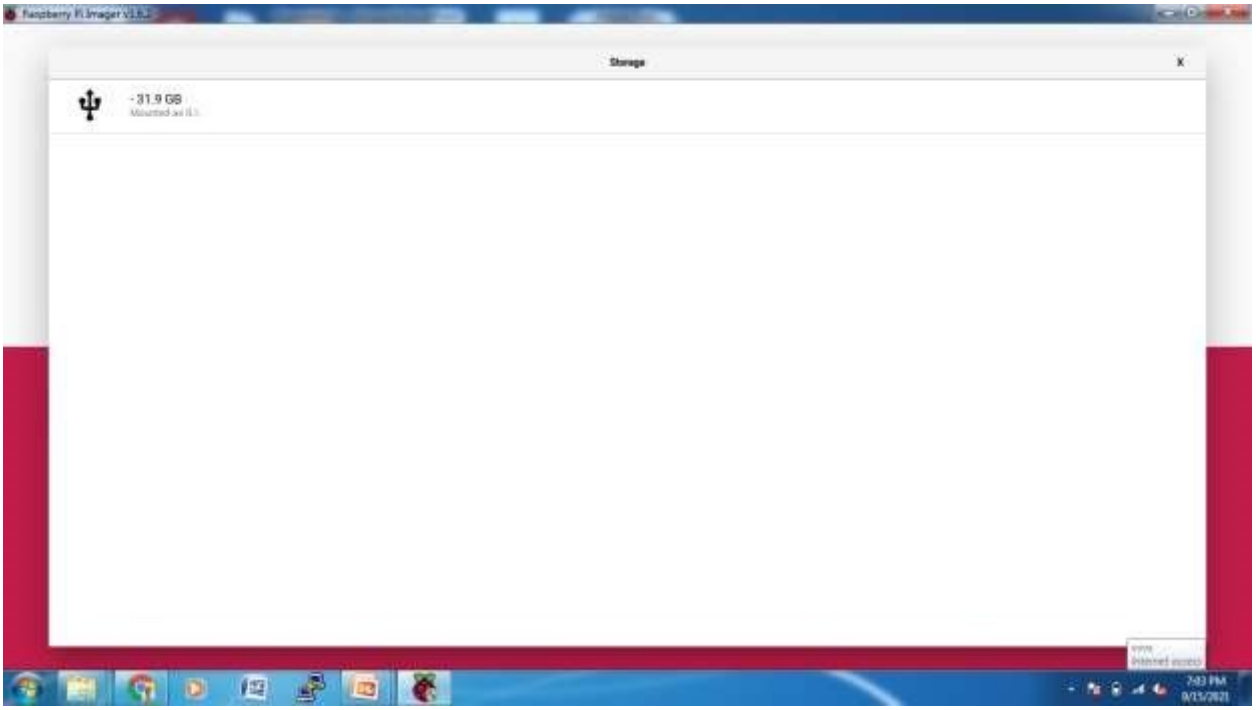
7) After opening the imager,the below window appears.Click on ‘CHOOSE OS’

8) Click on Use Custom and choose the operating system.



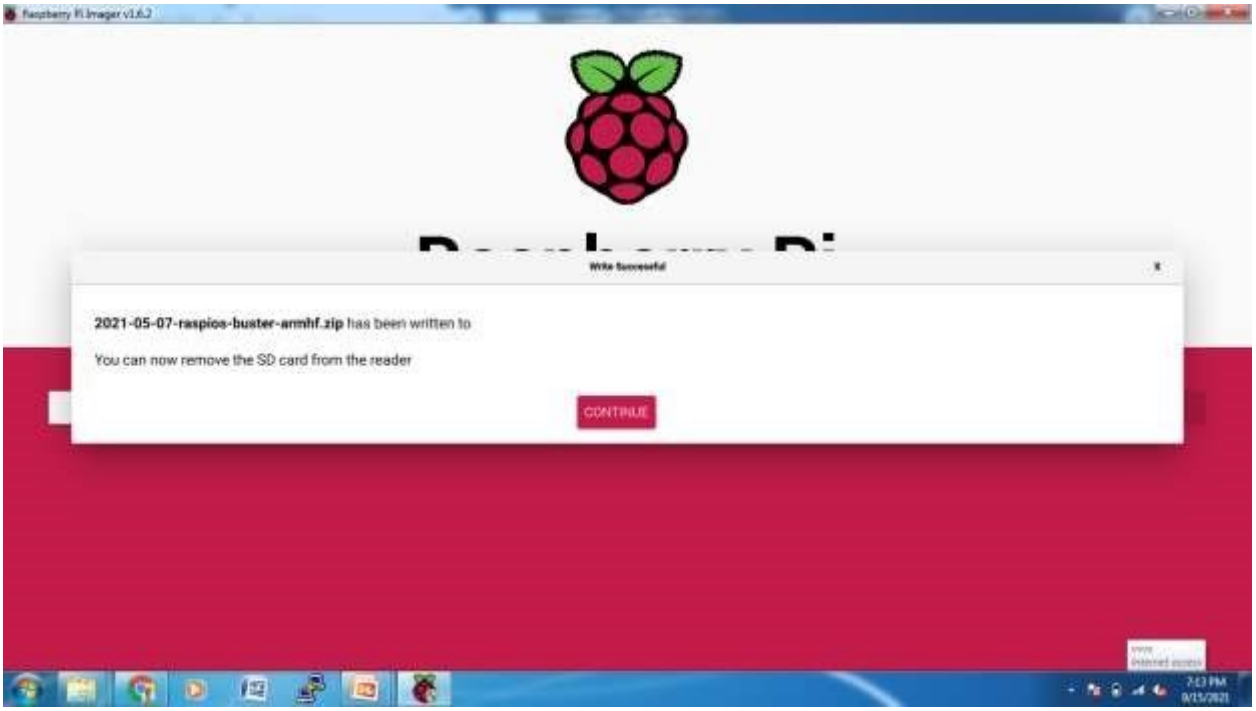


9) Now come back and 'Choose Storage'

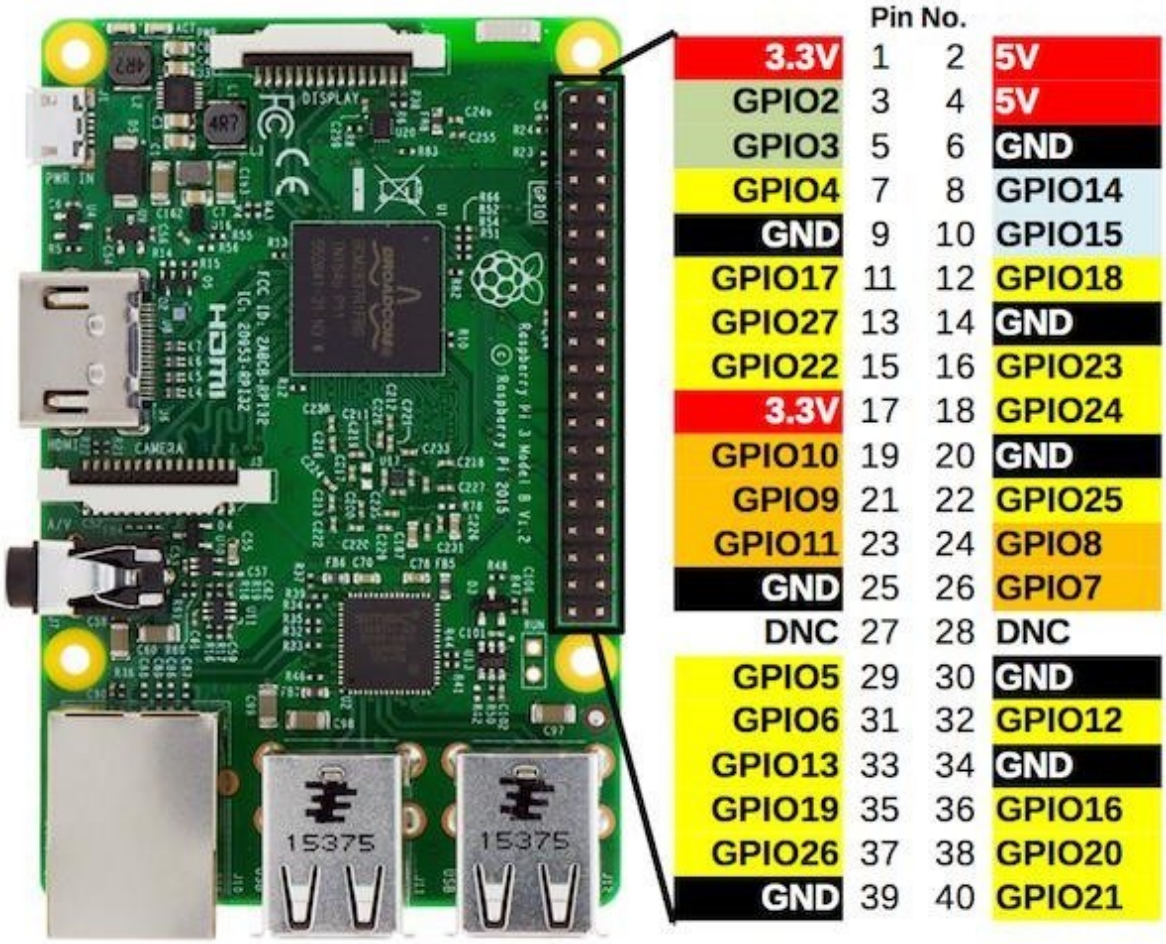


10) And lastly click on Write as shown in the below figure

11) The below window appears and thus Raspberry Pi is installed and Setup



RASPBERRY PI



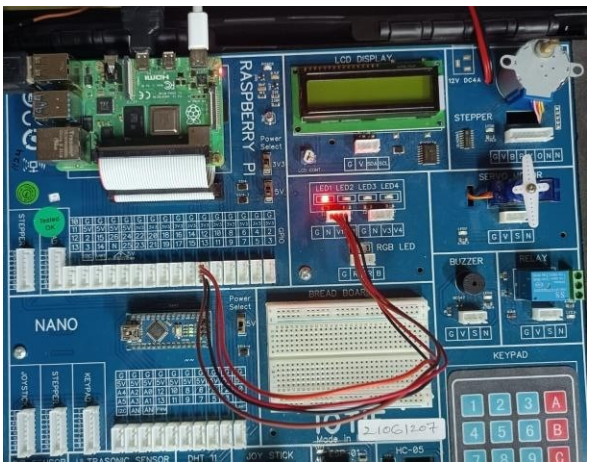
AIM: To implement a program to blink LED with raspberry pi.

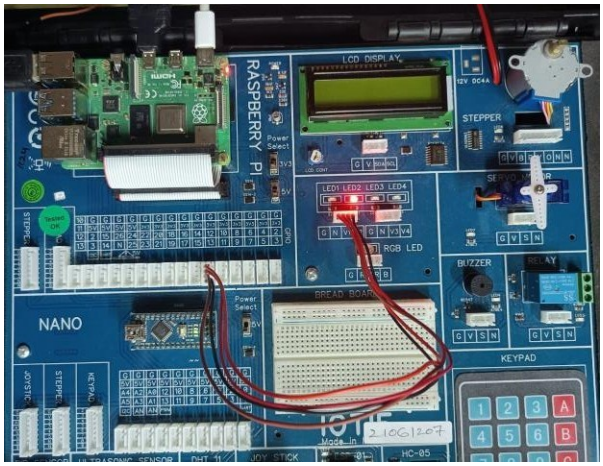
PROGRAM:

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BCM)
GPIO.setwarnings(False)

ledPinOne = 12
ledPinTwo = 13

GPIO.setup(ledPinOne, GPIO.OUT)
GPIO.setup(ledPinTwo, GPIO.OUT)

try:
```

```
    while True:
            GPIO.output(ledPinOne,HIGH)
            GPIO.output(ledPinTwo,LOW)
            time.sleep(100)
            GPIO.output(ledPinOne,LOW)
            GPIO.output(ledPinTwo,HIGH)
            time.sleep(100)
except KeyboardInterrupt:
    print("\nExiting Program\n")
    GPIO.cleanup()
    exit()
```

CIRCUIT:





OUTPUT: Demonstrated the set up and working of Raspberry Pi.

Sample Viva Questions:

1. Does Raspberry Pi need external hardware?
2. What is the difference between Aurdino and Raspberry Pi?
3. Does Raspberry pi have an internal memory?
4. How many GPIO pins are there in Raspberry Pi?
5. Which programming language is used in Raspberry Pi?

WEEK-11

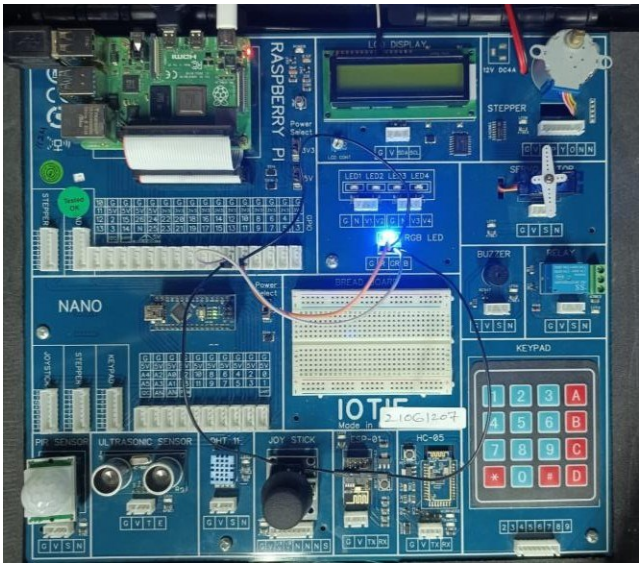
Interface RGB LED with Raspberry Pi to obtain different colours and brightnessusing PWM.

AIM: To interface RGB LED brightness using PWM pins with Raspberry Pi.

COMPONENTS REQUIRED: Raspberry Pi kit, system, connecting probes.

THEORY: Based on the high value that was send to a pin those corresponding color will be emitted out by RGB LED. Here we will observe the Green, Red, Blue, Pink colours simultaneously one after another. These colours will be emitted by the RGB LED continuously until the supply given to the LED is turned off. When any two or three pins are high, we will get another color because of combination of red, green, blue lights. here the Pi board behaves like a CPU to the monitor and it can perform operations related to CPU.

PROGRAM:




```
import RPi.GPIO as g

import time

g.setmode(g.BCM)

g.setwarnings(False)

ground = 10

red = 11

green = 12

blue  =  13 g.setup(red,

g.OUT) g.setup(green,

g.OUT)g.setup(blue,

g.OUT)

g.setup(ground, g.OUT)

g.OUTPUT(ground, g.LOW)p

= g.PWM(11,100)

q = g.PWM(12,100)r =

g.PWM(13,100)

p.start(0)

q.start(10)

r.start(50)
```

```
while True:
    for i in range(50):
        p.ChangeDutyCycle(i)
        q.ChangeDutyCycle(i)
        r.ChangeDutyCycle(i)
        time.sleep(0.5)
    for i in range(50):
        p.ChangeDutyCycle(100-i)
        q.ChangeDutyCycle(100-i)
        r.ChangeDutyCycle(100-i)
        time.sleep(0.5)
```

OUTPUT:

Hence a program to obtain different colours by using RGB LED which is being interfaced with the Raspberry Pi board is successfully executed and practically observed**.**

2.AIM: To interface RGB LED with Raspberry Pi to obtain different colors through user input. PWM.

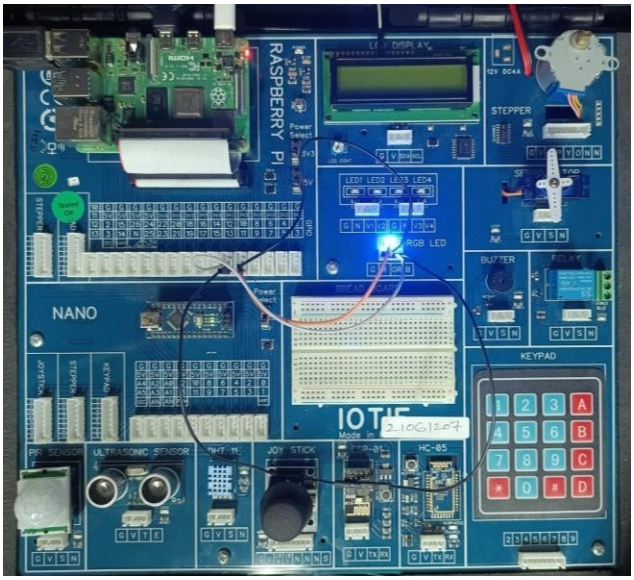
COMPONENTS REQUIRED: Raspberry Pi kit, system, connecting probes.

THEORY: The Raspberry Pi's GPIO gives either 3.3 volts as maximum or zero volts, so the output is a square wave and in order to vary the brightness, we cannot get the voltage between 0-3.3 volts.Therefore, here we use the concept of duty cycle that is the on and off time of the signal and vary the brightness accordingly. The thing which is to be noted here is that with the variation of the duty cycle the brightness of a led can be varied with the help of Pulse Width Modulation(PWM).constantly changing the duty cycle changes the T-on and T-off time and thus we can change the brightness of the led.

PROGRAM:

```
import RPi.GPIO as g
import time
g.setmode(g.BCM)
g.setwarnings(False)
ledpinone = 11
ledpintwo = 12
```

```
ledpinthree = 13
g.setup(10, g.LOW)

g.setup(ledpinone, g.OUT)

g.setup(ledpintwo, g.OUT)

g.setup(ledpinthree, g.OUT)

while True:

    color = input() if

    color == 'red':

        g.OUTPUT(ledpinone, g.HIGH)

        g.OUTPUT(ledpintwo, g.LOW)

        g.OUTPUT(ledpinthree, g.LOW)

    elif color == 'green':

        g.OUTPUT(ledpinone,        g.LOW)

        g.OUTPUT(ledpintwo,       g.HIGH)

        g.OUTPUT(ledpinthree, g.LOW)

    elif color == 'blue':

        g.OUTPUT(ledpinone, g.LOW)

        g.OUTPUT(ledpintwo, g.LOW)

        g.OUTPUT(ledpinthree, g.HIGH)
```



OUTPUT: Hence a program to obtain different brightnessof RGB LED using PWM , which is being interfaced with the Raspberry Pi board is successfully executed and practically observed.

Sample Viva Questions:

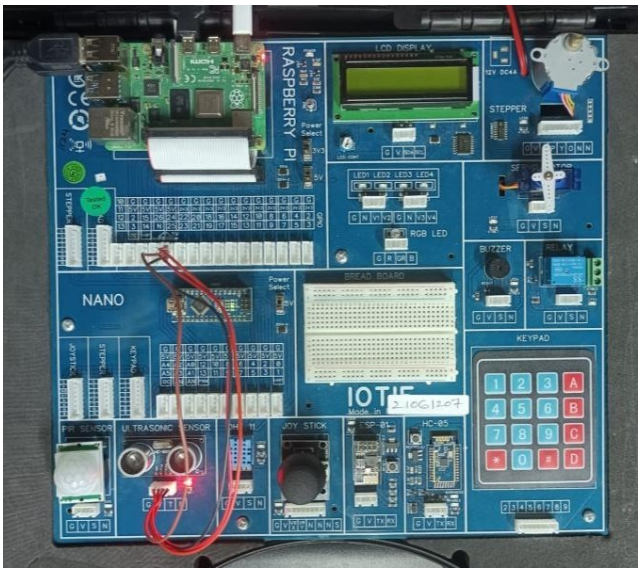
1. What is PWM?
2. What is color code for red?
3. What does the 4 pins for RGB Led indicate?
4. How do we obtain white color in RGB LED?
5. How is intensity changed in PWM method?
6. What is duty cycle?

WEEK-12

Interface an ultrasonic sensor with Raspberry pi to print distance readings on the monitorwhen the sensor changes its position. b) Reading the data from an analog sensor with Raspberry using Arduino serial port or ADC MCP3208 using SPI.

1.AIM: To interface an ultrasonic sensor with Raspberry pi to print distance readings on the monitorwhen the sensor changes its position.

COMPONENTS REQUIRED: Raspberry Pi kit, monitor, connecting probes



THEORY:

An ultrasonic sensor is an instrument that measures the distance to an object using ultrasonic sound waves. An ultrasonic sensor uses a transducer to send and receive ultrasonic pulses that relay back information about an object's proximity. The working principle of this module is simple. It sends an ultrasonic pulse out at 40kHz which travels through the air and if there is an obstacle or object, it will bounce back to the sensor. By calculating the travel time and the speed of sound, the distance can be calculated.

PROGRAM:

```
import RPi.GPIO as g

import time

def distance(trigpin, echopin):

    g.output(trigpin, True)

    time.sleep(0.0001)

    g.output(trigpin, False) while

    g.input(echopin) == 0:

      pulse_start = time.time() while
```

```
    g.input(echopin) == 1:pulse_end
                = time.time()
      try:
       pulse_duration = pulse_end - pulse_start
    except:
       print('Calibrating')
       return -1
    distance = pulse_duration * 17150
    distance = round(distance + 1.15, 2)
    return distance

g.setmode(g.BCM)
trigpin = 24
echopin = 25
g.setup(trigpin, g.OUT)
g.setup(echopin, g.IN)

while True:
   dist = distance(trigpin, echopin) print('Measured
   distance = {}cm'.format(dist))time.sleep(0.01)
```

OUTPUT:

The distance is measured continously and is displayed in the serial window until interrupted by the keyboard.

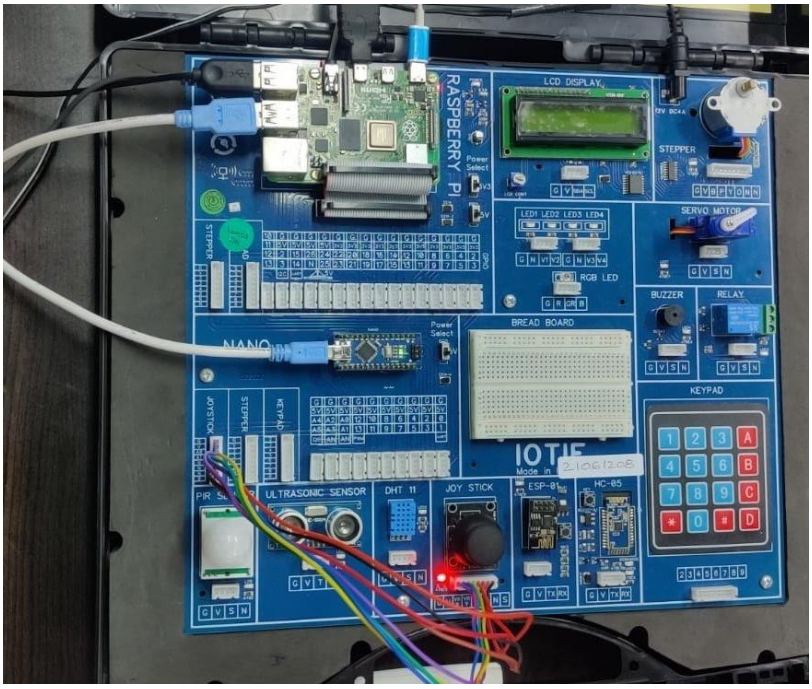
2. AIM: **To read the data from an analog sensor (joystick) with Raspberry using Arduino serial port.**

THEORY:

"**Joystick**" is simply a game controller. Because everybody has seen it while playing video games. But, apart from gaming, it can wide range of applications in electronics. Actually, the joystick is the combination of two potentiometers for the X and Y Plane. It gives analog data to the Arduino by reading the voltage from

the potentiometer and it is given to raspberry pi. So that we can get the readings of joystick by using raspberry pi.We cannot give the joystick directly to raspberry pi as it won't allow analogdata to run.Thisanalog value get changes as we move the joystick shaft.

PROGRAM:




```
# ## On Arduinoint
VRx = A0;
int VRy = A1;int
SW = 2;
int xposition = 0;int
yposition = 0;int
SW_state = 0 int
mapx = 0
int mapy = 0; void
setup() {
   Serial.begin(9600); pinMode(VRx,
   INPUT); pinMode(VRy, INPUT);
   pinMode(SW, INPUT_PULLUP);
}
void loop()
{
   xposition    =    analogRead(VRx);
   yposition    =    analogRead(VRy);
   SW_state = digitalRead(SW);
   mapx = map(xposition, 0, 1023, -512, 512);
   mapy = map(yposition, 0, 1023, -512, 512);
   Serial.print("X:");
   Serial.print(mapx);
   Serial.print("Y:");
   Serial.print(mapy);
   Serial.print("/Button");
```

```
  Serial.println(SW_state);
  delay(100);
}

# ON Raspberry pi
import serial import
time
if _name_ == "_main_":
  ser = serial.Serial('/dev/ttyUSB0', 9600, timeout=1)
  ser.reset_input_buffer()

while True:
  if ser.in_waiting>0:
    line = ser.readline().decode('utf-8').rstrip()
    print(line)
```

OUTPUT: (will look similar to this)

Sample Viva Questions:

1. What is an IR sensor?
2. List different types of Analog and digital sensors.
3. What are the terminals of a joystick?
4. Explain the working of Ultrasonic sensor?
5. What is the range and resolution of DHT sensor?
6. What is the difference between ultrasonic and PIR sensors?

WEEK-13

Post/read the data to/from the cloud via MQTT broker with a Raspberry Pi.

AIM: To post the data to the cloud via MQTT broker with a Raspberry Pi.

COMPONENTS REQUIRED: Raspberry Pi kit, monitor, connecting probes

THEORY:

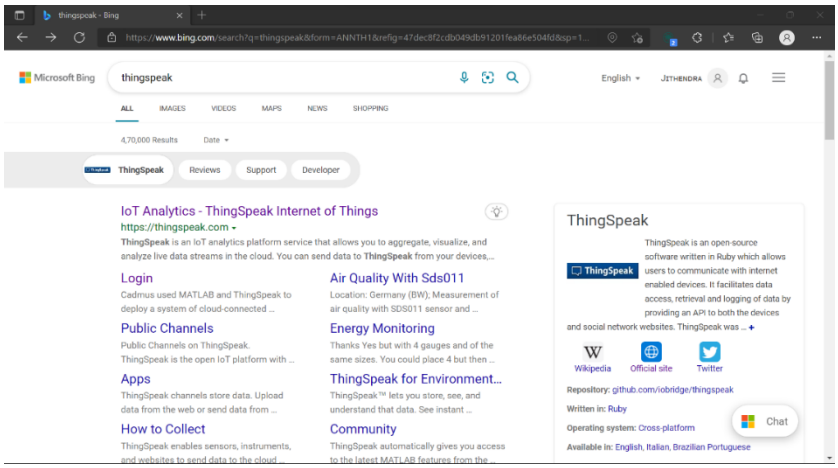
The Raspberry Pi is a low cost, credit-card sized computer that plugs into a computer monitor or TV, and uses a standard keyboard and mouse. DHT11 is a Digital Sensor consisting of a Negative Temperature Coefficient Temperature Sensor, a Resistive-type Humidity Sensor and an 8-bit Microcontroller to convert the analog signals from these sensors and produce a Digital Output. The output from the DHT11 Sensor is Digital.

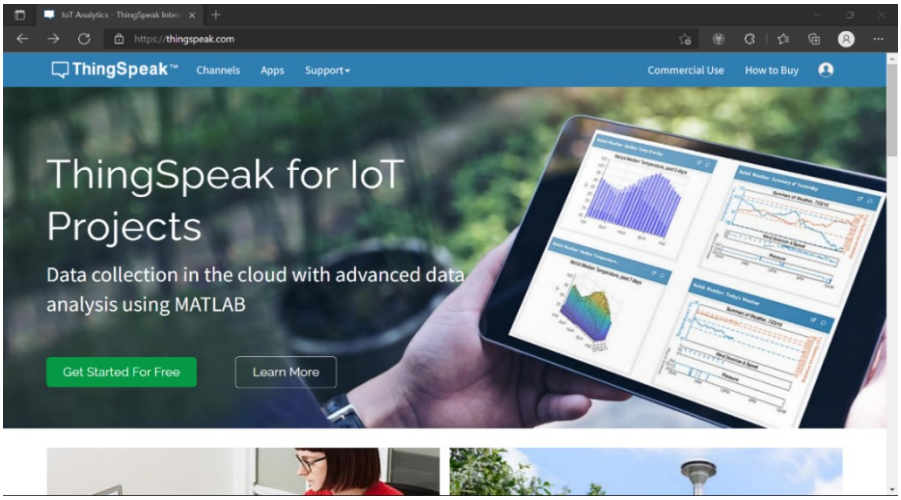
ThingSpeak is an IoT analytics platform service that allows you to aggregate, visualize, and analyze live data Streams in the cloud. Data can be sent to ThingSpeak from devices, create instant visualization of live data, and send alerts.

Creating a channel in ThingSpeak Cloud:

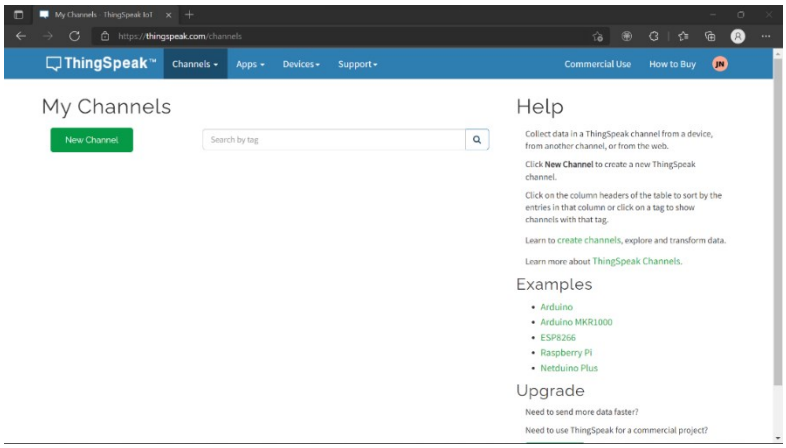
Go to the search engine of your personal computer and search for the official website of ThingSpeak.

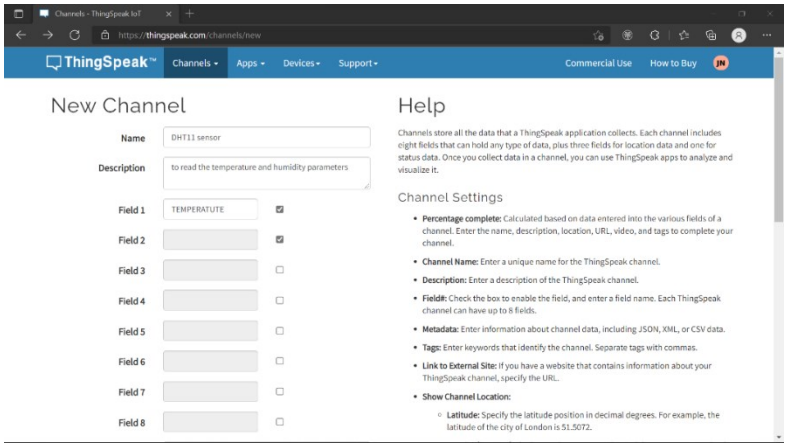
1.Click on the website and signup for an account by clicking on the top right corner of the page as shown in below figure.





2After signing up, click New Channel to create a new ThingSpeak channel.

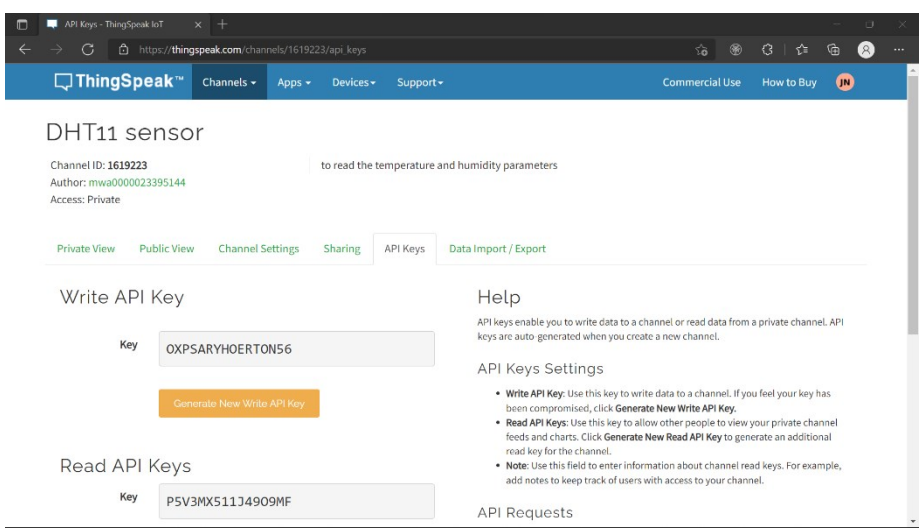


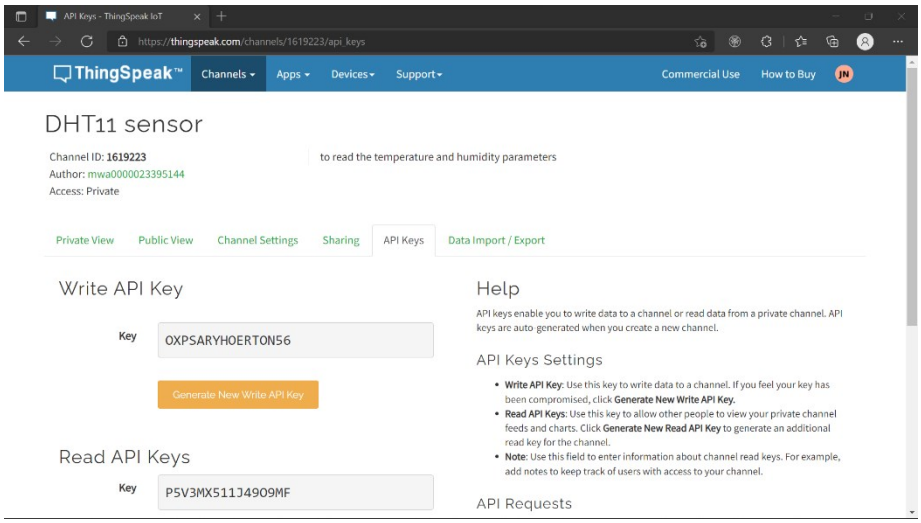


3.Give the name of the channel, its description and field name as Temperature and Humidity as shown in below figure and save the channel.

After saving the channel go to the API Keys menu which enables to write data to a channel or read data from a private channel. Copy the Write API Key as the data is written to the channel

‘





Copy the Write API Key as the data is written to the channel

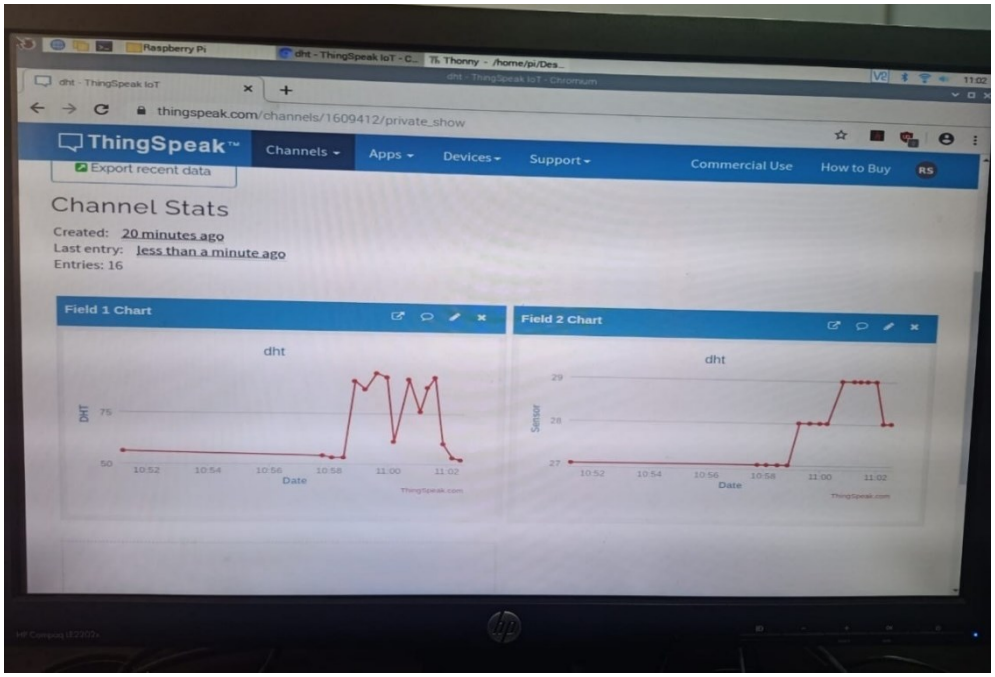
CODE:

```
importurllib.request

import time

import RPi.GPIO as GPIO

import Adafruit_DHT

writeAPIKey = "8CMZMVX6CQELK7KG" # enter write API key here

baseURL = "https://api.thingspeak.com/update?api_key={}".format(writeAPIKey)
sensor = Adafruit_DHT.DHT11

sensorPin = 26

GPIO.setmode(GPIO.BCM) # Using BCM numbering

try:

 while True:

humidity, temperature = Adafruit_DHT.read_retry(sensor, sensorPin)

 if humidity is not None and temperature is not None:

 # Format the readings to two decimal places

 humidity = '%.2f' % humidity

                    temperature = '%.2f' % temperature

                    # Sending the readings to thingspeak
```

```
conn = urllib.request.urlopen(baseURL + '&field1={}&field2={}'.format(humidity, temperature))
print(conn.read())

 # Closing the connection

conn.close()

time.sleep(20) # wait 20 seconds before uploading next reading except
KeyboardInterrupt:

GPIO.cleanup()

exit()
```

OUTPUT:

The DHT11 sensor is interfaced with Raspberry Pi and the readings of Temperature and Humidity are successfully posted on the ThingSpeak cloud and also observed the variations of DHT11 sensor in the field charts as shown in below figure.



2. AIM: To implement a program to send real-time sensor data to smart phone using RaspberryPi onboard Bluetooth.

THEORY: On the raspberryPi board monitor we are having a bluetooth connection symbol at the bottom right corner click on 'add a device' and check for the device or mobile bluetooth device name in which mobile bluetooth is already switched ON. Next pair the bluetooth devices by clicking on the 'allow' and 'pair' button/options. After these steps the Pi board and the mobile device are connected to the bluetooth and they can transfer data in between them. BY placing an object in front of an ultrasonic sensor it can calculate the distance corresponding to it and it will be printed on the monitor screen and the mobile device screen which is connected through bluetooth. On the mobile the distance measured by the sensor can be observed by opening 'Bluetooth terminal' connection.

Here's what you'll need to do to install it:

- Open Terminal and type **sudo apt-get install bluetooth bluez blueman**. Press enter.
- Once the packages have downloaded and installed, type **sudo reboot**.
- To access this menu, on the Raspberry Pi desktop click Menu, located in the upperleft corner of the screen, scroll down to **Preferences** with your cursor and click **Bluetooth Manager**. From there, you can pair to any nearby devices, or you can make your Raspberry Pi discoverable so you can pair your phone to it from your phone's Bluetooth settings.

How to setup Bluetooth on Raspberry Pi:

- From the Raspberry Pi desktop, open a new Terminal window.
- Type **sudo bluetoothctl** then press enter and input the administrator password (thedefault password is **raspberry**).
- Next, enter **agent on** and press enter. Then type **default-agent** and press enter.
- Type **scan on** and press enter one more time. The unique addresses of all the Bluetooth devices around the Raspberry Pi will appear and look something like analphanumeric **XX:XX:XX:XX:XX:XX**. If you make the device you want to pair

  discoverable (or put it into pairing mode), the device nickname may appear to the right of the address. If not, you will have to do a little trial and error or waiting to findthe correct device.

- To pair the device, type **pair [device Bluetooth address]**. The command will looksomething like pair **XX:XX:XX:XX:XX:XX**.
- If you're pairing a keyboard, you will need to enter a six-digit string of numbers. You will see that the device has been paired, but it may not have connected. To connect thedevice, type **connect XX:XX:XX:XX:XX:XX**.

PROGRAM:

```
import bluetooth
import time

import RPi.GPIO as g


g.setmode(g.BCM)

trigpin = 24
```

```
echopin = 25
g.setup(trigpin, g.OUT)
g.setup(echopin, g.IN) host
= " "
port = 1
server = bluetooth.BluetoothSocket(bluetooth.RFCOMM)try:
   server.bind((host, port)) print("bluetooth
   binding succesful")
except:
   print('Bluetooth binding unsuccesful')
   server.listen(1)
   client, address = server.accept()
   print('connection received from', address)
   print('client', client)

   def distance(trigpin, echopin):
      g.output(trigpin, True)
      time.sleep(0.0001)
      g.output(trigpin, False) while
      g.input(echopin) == 0:
        pulse_start = time.time() while
       g.input(echopin) == 1:pulse_end
                    = time.time()
         try:
          pulse_duration = pulse_end - pulse_start
      except:
          print('Calibrating')
          return -1
      distance = pulse_duration * 17150
      distance = round(distance + 1.15, 2)
```

```
    return distance

while True:
    dist = distance(trigpin, echopin) print('measured
    distance = {} cm'.format(dist))client.send('\n
    measured distance') client.speed(str(dist))
    time.sleep(2)
```

OUTPUT: Hence a program for sending the real-time sensor data values to a smartphone using Raspberry Pi on boardbluetooth has been successfully executed.

Sample Viva Questions:

1. What is a MTTQ broker?
2. Does MTTQ support security.
3. Give use cases of sending data through cloud.
4. What is the difference between sending data through Bluetooth and cloud?
5. What is API?

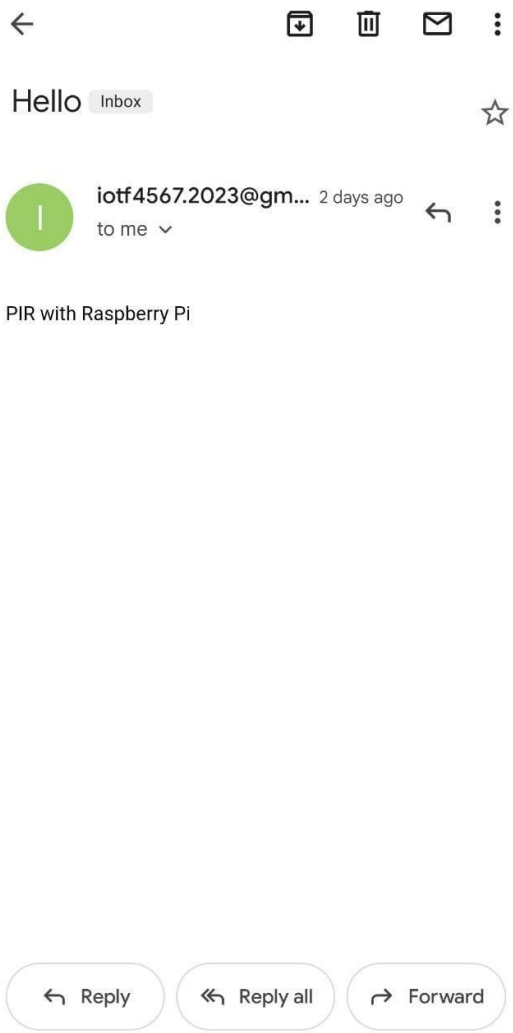
WEEK-14

AIM: To implement an intruder alert system that alerts through email.

PROGRAM:

```
import smtplib
import time
from time import sleep
import RPi.GPIO as GPIO
from sys import exit

from_email = '<my-email>' receipients_list
= ['<receipient-email>']cc_list = []
subject = 'Hello'
message = 'PIR with Raspberry Pi'
username = '<Gmail-username>'
password = '<password>' server =
'smtp.gmail.com:587'

GPIO.setmode(GPIO.BCM)
GPIO.setwarnings(False)

GPIO.setup(11, GPIO.IN)


def sendmail(from_addr, to_addr_list, cc_addr_list, subject, message, login, password,
smtpserver):
   header = 'From: %s \n' % from_addr header +=
   'To: %s \n' % ','.join(to_addr_list)header += 'Cc:
   %s \n' % ','.join(cc_addr_list)header += 'Subject:
   %s \n \n' % subject message = header + message

   server = smtplib.SMTP(smtpserver)
   server.starttls()
   server.login(login, password)
   problems = server.sendmail(from_addr, to_addr_list, message)
   server.quit()

while True:try:
       if (GPIO.input(11) == 1):
          sendmail(from_email, receipients_list, cc_list, subject, message, username, password,server)
       sleep(.01)
    except KeyboardInterrupt:
       exit()
```

OUTPUT:



Sample Viva Questions:

1. What are intruder alarm systems?
2. Give use cases of this program.
3. What components are used in developing an alarm system using Raspberry?

WEEK-15

AIM: To implement a program to send an email with Raspberry Pi.

PROGRAM:

```
import smtplib

from time import sleep

import RPi.GPIO as GPIO

from sys import exit


from_email = '<my-email>' receipients_list

= ['<receipient-email>']cc_list = []

subject = 'Hello: Message from Raspberry Pi'
message = 'Input given by the user'

username = '<Gmail-username>'

password = '<password>'

server = 'smtp.gmail.com:587'


inp=int(input())

def sendmail(from_addr, to_addr_list, cc_addr_list, subject, message, login, password,
smtpserver):

        header = 'From: %s \n' % from_ addr header +=

        'To: %s \n' % ','.join(to_addr_list)header += 'Cc:

        %s \n' % ','.join(cc_addr_list)header += 'Subject:

        %s \n \n' % subject message = header + message


        server = smtplib.SMTP(smtpserver)

        server.starttls()

        server.login(login, password)

        problems = server.sendmail(from_addr, to_addr_list, message)

        server.quit()
```

```
while True:try:
        if (inp == 1):
            sendmail(from_email, receipients_list, cc_list, subject, message, username, password,
server)
        sleep(.01)
    except KeyboardInterrupt:
        exit()
```

OUTPUT:

Sample Viva Questions:

1. What is the protocol used to send an email through Raspberry Pi?
2. What are the components of SMTP protocol?
3. Which Programming language is used in setting up SMTP protocol?